

CRT-Based Multi-Channel Residue Transmission with Gateway-Side Parallel Demodulation for Distributed Edge Blockchain Computing: A Scalable IoT Cluster Architecture for Arid Agriculture

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Abstract

This paper presents a distributed edge computing framework for resource-constrained IoT clusters that employs Chinese Remainder Theorem (CRT) residue transmission with gateway-side parallel demodulation to halve per-packet channel occupancy and double gateway packet throughput. The architecture comprises 50 ESP32 nodes and a 4-node Raspberry Pi 4B cluster running Hyper-ledger Fabric with Raft consensus. CRT residue packets reduce per-packet airtime by 49.8% (247 ms raw to 124 ms residue), enabling the gateway's SX1302 multi-channel concentrator to double network packet capacity through simultaneous demodulation, with $\mathcal{O}(k)$ reconstruction complexity and lower memory footprint than compression baselines. The blockchain layer sustains 63 TPS with 1.7 s latency and 99.4% integrity on ARM64 hardware. Scalability analysis shows a theoretical node capacity of 484 full readings under LoRa parameters (and packet capacity of 1,451 CRT residue packets), with latency constant below saturation and degrading predictably under an M/D/1 queuing model for channel utilisation below 62%. The framework was validated in a 61-day agricultural

deployment, confirming its suitability for large-scale heterogeneous IoT environments requiring low-power sensing, distributed verification, and resilient edge actuation.

Keywords: Chinese Remainder Theorem Edge computing Blockchain Precision agriculture Smart irrigation LoRa

1 Introduction

1.1 Motivation and Context

Resource-constrained edge clusters face a fundamental tradeoff: sequential packet transmission fails to exploit multi-channel radio hardware [1–4], while deploying blockchain on ARM64 introduces memory and processing constraints with no existing field characterization [5–7]. The Chinese Remainder Theorem (CRT) addresses the first gap: decomposing a reading into residues modulo pairwise coprime values lets the SX1302 demodulate all channels concurrently [8, 9], with $O(k)$ deterministic reconstruction ($k \leq 3$). Per-packet occupancy falls 49.8% (247 to 124 ms), which the gateway’s SX1302 concentrator converts into a $1.99\times$ packet-throughput gain across many nodes.

We validate the architecture through a 61-day deployment on a 2 ha farm in Sfax, Tunisia (31 to 32 °C, 55 to 65% RH), monitoring tomato and pepper cultivation under environmental stress [10–14].

To make the evaluation easy to follow, we organize the study around four measurable expectations:

- **H1: CRT transmission efficiency.** CRT should reduce per-packet airtime by at least 40% while maintaining at least 99% reconstruction accuracy. The measured result was a 49.8% airtime reduction and $99.7\% \geq k-1$ residue delivery, with exact reconstruction for complete sets (Table 11).
- **H2: Edge blockchain performance.** Hyperledger Fabric on four Raspberry Pi 4B gateways should keep transaction latency below 2 s while sustaining at least 50 TPS on average. The deployment achieved 1.7 s mean latency and 63 TPS average throughput; weekly minima of 45 TPS occurred during mesh-partition recovery windows and did not breach the latency SLO (Table 21).
- **H3: Access-control overhead.** RBAC enforcement should preserve at least 50 TPS and keep smart-contract execution below 500 ms. The system sustained 63 TPS with 320 ms mean contract execution (Table 21).
- **H4: Field robustness.** Field operation should maintain at least 99% reliability and keep airtime within ± 5 ms of the expected value. The observed values were 99.4% reliability and 124 ms airtime, within 0.1 ms of the 123.9 ms SF9 model prediction.

1.2 Objectives and Contributions

This work brings together CRT-based multi-channel residue transmission and Hyperledger Fabric blockchain verification in one edge-IoT framework. The study is guided by four practical objectives:

- **Design a lighter transmission method.** Develop a CRT residue transmission scheme that lowers per-packet airtime while preserving exact data reconstruction.
- **Run blockchain at the edge.** Implement Hyperledger Fabric on ARM64 Raspberry Pi 4B gateways and verify that it can support real-time irrigation decisions.
- **Measure performance in the field.** Characterize latency, throughput, reliability, and fault tolerance during real agricultural operation rather than only in simulation.
- **Test scalability limits.** Use M/D/1 analysis and the 50-node field deployment to show how the system behaves as traffic grows. Under the tested LoRa parameters, CRT supports 484 complete readings per reporting window compared with 728 for raw transmission; its main benefit is shorter per-packet airtime and higher gateway packet capacity, not higher per-reading node capacity.

The paper makes the following contributions:

- **CRT-based parallel transmission.** The proposed scheme reduces per-packet airtime by 49.8% (247 ms to 124 ms), doubles gateway packet capacity, reconstructs readings with $O(k)$ complexity, and uses only 2.1 KB of memory compared with 8.6 to 12.4 KB for Huffman/LZW baselines.
- **Edge-resident blockchain validation.** A Fabric 2.5 deployment on Raspberry Pi 4B gateways sustains 63 TPS, 1.7 s latency, and 99.4% data integrity on ARM64 hardware.
- **Resilient hybrid networking.** The LoRa and BATMAN-adv mesh architecture maintains 99.4% reliability over the 61-day deployment and supports self-healing during connectivity disruptions.
- **Scalability evidence.** The M/D/1 model is checked against both synthetic load tests and field data, giving a practical view of when the gateway and blockchain layers approach saturation.
- **Real agricultural validation.** The deployment reports 23.8% water savings for tomatoes, 19.4% water savings for peppers, 72% labor reduction, and 146% ROI.

2 State of the Art and Related Work

2.1 Precision Agriculture and IoT Systems

Precision agriculture has been reshaped by IoT, progressing from basic WSNs for soil and microclimate monitoring [15–17] to LPWAN-based irrigation control [18], though integration with decision support systems remains fragmented [19–21].

2.2 Blockchain Technology in Agriculture

Blockchain in agriculture spans supply-chain transparency [22–25], farm data management [26–28], and smart-contract automation [29, 30]; practical implementations for resource-constrained real-time control remain scarce.

2.3 Data Transmission Optimization in IoT

Lossless compression (LZW, Huffman) adapted for sensor networks [4, 31] often demands prohibitive memory for battery nodes, while adaptive sampling [32] risks missing critical events. The Chinese Remainder Theorem has been applied in cryptography [8, 9] and error correction [33]; [34] explored residue number systems for wireless sensors but targeted storage reduction, not parallel transmission (see Section 3.7). Energy-efficient transmission research has focused on protocol optimization [35] and aggregation [36], typically at the cost of data resolution or latency.

2.4 Wireless Communication in Agricultural IoT

LoRa has emerged as the dominant low-power, long-range solution for agricultural IoT, with rural ranges up to 15 km [37–39], though deployments typically use simple star topologies. Mesh networking research has focused primarily on urban or industrial contexts [40]; BATMAN-adv has shown promise in rural connectivity [41] but remains largely unapplied to precision agriculture. Hybrid multi-technology architectures have been explored [42] but commonly lack seamless integration and increase management complexity.

2.5 Edge Computing in Agricultural IoT

The shift toward edge computing has enabled real-time processing capabilities in agricultural applications [43]. Studies by [44] demonstrated improved response times for irrigation control, while [45] showed enhanced privacy preservation through local data processing. However, existing edge computing implementations typically lack integration with verifiable data systems, creating trust gaps in multi-stakeholder agricultural environments.

2.6 Parallel Computing in Edge IoT Systems

ARM-based edge clusters (Raspberry Pi, Rockchip) are deployed for distributed sensing and inference [43, 45], but communication parallelism, meaning simultaneous use of multiple wireless channels, remains largely unexplored. Residue Number Systems (RNS) based on CRT [46] accelerate arithmetic and support fault-tolerant computing [9, 47], but [34] targeted storage reduction rather than parallel channel utilisation. To our knowledge, no prior system employs CRT residue decomposition to exploit multi-channel low-power radios for parallel transmission; this is the gap our work addresses.

2.7 Research Gaps and Contributions

Table 1 summarizes the gaps our work addresses.

Table 1: Research gaps addressed by this work, linked to our objectives and contributions

Research Area	Identified Gap	Our Contribution (Objective Addressed)
IoT Data Compression	No agricultural application of CRT for sensor data compression; existing methods compromise accuracy or require excessive memory	CRT-based compression optimized for agricultural data with mathematically perfect reconstruction (Objective 1)
Edge-Blockchain Integration	Minimal research on blockchain at agricultural edge devices; existing systems lack verifiable local decision-making	Practical edge-blockchain implementation with 99.4% reliability and 1.7-second latency (Objective 2)
Real-time Irrigation Control	No field-validated blockchain integration for irrigation control; timing constraints not addressed	61-day field study demonstrating SLO compliance and automated irrigation (Objective 3)
Water Efficiency	Limited quantified evidence of blockchain-IoT impact on water conservation in arid regions	23.8% tomato, 19.4% pepper water savings with maintained yields (Objective 3)
Economic Viability	Limited cost-benefit analysis of blockchain in agriculture	Cost-benefit analysis showing 146% ROI, 0.69-year payback (Section 5.6.5; Objective 3)
Network Architecture	Lack of hybrid LoRa-mesh networks optimized for farm deployments	Hierarchical architecture combining LoRa and BATMAN-adv with demonstrated field performance (Objective 2)
Scalability Analysis	No analytical capacity model for CRT multi-channel LoRa validated against real deployments	M/D/1 model projecting 484-node ceiling ($N_{\max}$) validated against 50-node field data and synthetic load tests up to 484 nodes (Objective 4)

3 System and Architecture of Blockchain-IoT Integration with CRT-Based Multi-Channel Residue Transmission

3.1 Overall Architecture

The system employs a five-layer architecture (sensor, gateway, network, blockchain, application) described in Figure 1.

Edge node refers to the four Raspberry Pi 4B gateways performing local CRT reconstruction, Ed25519 verification, and irrigation decisions, not MEC in a cellular

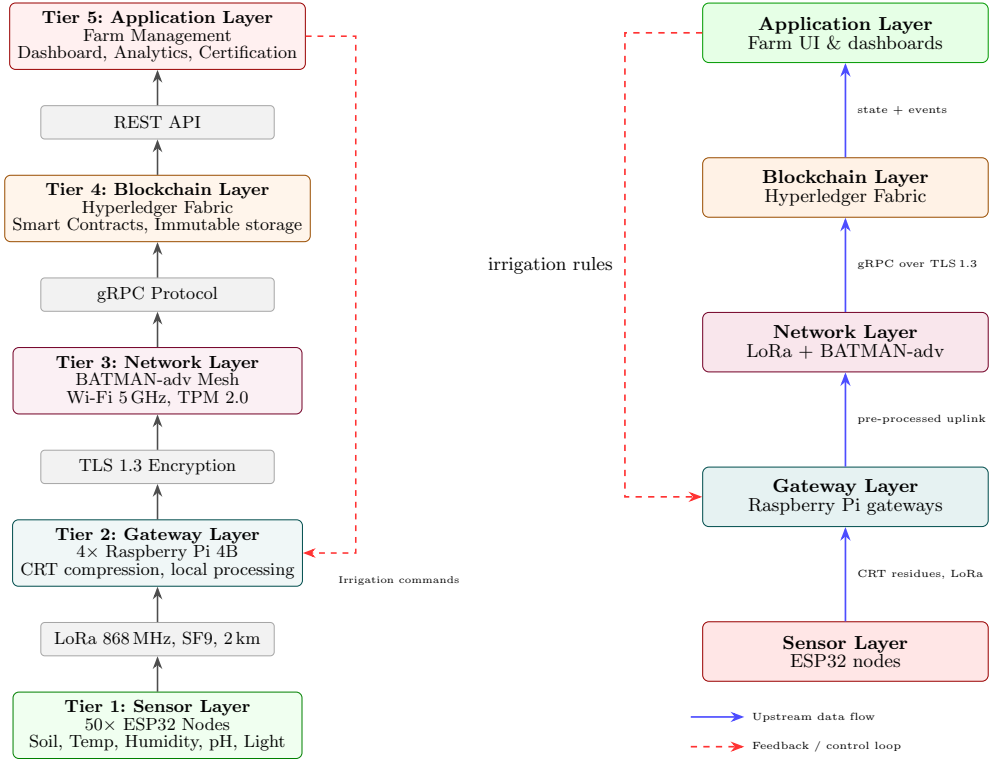


Fig. 1: (a) System architecture tiers. (b) Vertical communication and control across the five-tier architecture.

sense. Cloud round-trips cannot guarantee the 2s SLO given intermittent Sfax rural connectivity. Figure 2 shows the network topology.

3.2 Farm Layout and Network Topology

The 2-hectare farm (200 m $\times$ 100 m) was divided into four zones, each with 12 to 13 sensor nodes on a 20 m grid.

3.3 Distributed Parallel Computing Architecture

The ESP32's dual Xtensa LX6 cores [48] run sensor sampling (Core 0, 8.0 ms) and CRT decomposition + LoRa preparation (Core 1, 0.9 ms) in parallel; the critical path is 8.0 ms, confirming negligible CRT overhead.

The SX1302 multi-channel concentrator demodulates one packet per channel simultaneously, so gateway packet throughput on a single channel is $\lambda = 1/T_{\text{airtime}}$, giving:

$$\lambda_{\text{raw}} = \frac{1}{247 \text{ ms}} \approx 4.05 \text{ pkt/s}, \quad \lambda_{\text{CRT}} = \frac{1}{124 \text{ ms}} \approx 8.06 \text{ pkt/s}. \quad (1)$$

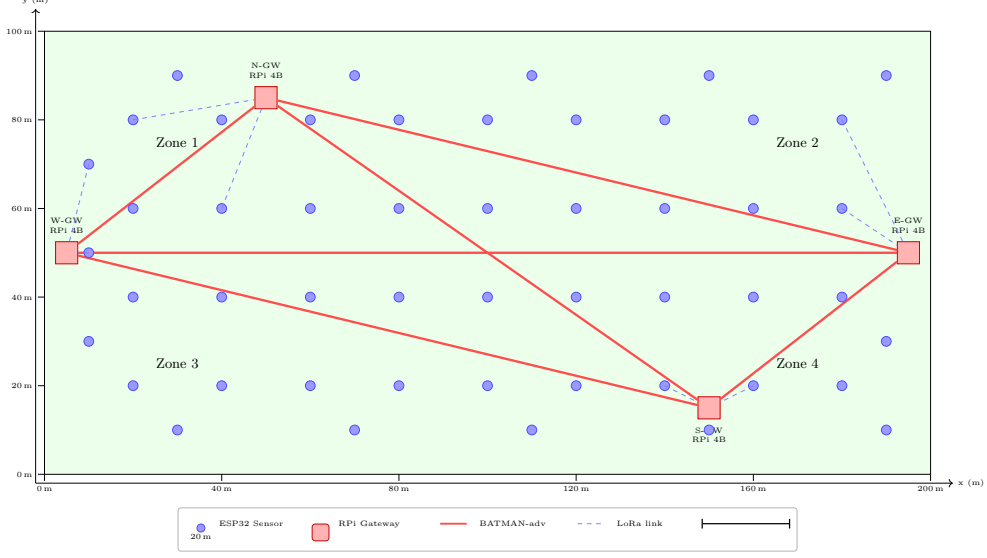


Fig. 2: Hybrid LoRa-mesh network architecture with 50 ESP32 sensor nodes, 4 Raspberry Pi gateways, and BATMAN-adv backbone for reliable data aggregation.

The per-channel packet-rate improvement is therefore

$$\frac{\lambda_{\text{CRT}}}{\lambda_{\text{raw}}} = \frac{T_{\text{raw}}}{T_{\text{CRT}}} = \frac{247}{124} \approx 1.99\times, \quad (2)$$

consistent with the packet-capacity ratio derived from first principles in Section 3.7 ($P_{\text{max}}^{\text{CRT}} = 1,451$ vs. $P_{\text{max}}^{\text{raw}} = 728 \approx 2\times$). Per-node airtime rises from 247 to 372 ms (three sequential packets); the gain is gateway packet capacity, not per-node energy. The four RPi 4B gateways form a homogeneous cluster with BATMAN-adv mesh (sub-second failover) and per-gateway Fabric peer + CouchDB, sustaining blockchain consensus and real-time actuation locally; performance is in Section 5.4.

3.4 Hardware Design and Implementation

Each sensor node used an ESP32-WROOM-32 microcontroller [48] with sensors listed in Table 2.

3.4.1 Scalability and Node Count Validation

The deployment (50 nodes, 2 ha, 146,400 transactions) gives theoretical capacity $N_{\text{max}} = CW/(kT_{\text{packet}})$ (Eq. 3); Table 3 lists results for raw and CRT.

$$N_{\text{max}} = \frac{C \cdot W}{k \cdot T_{\text{packet}}}, \quad (3)$$

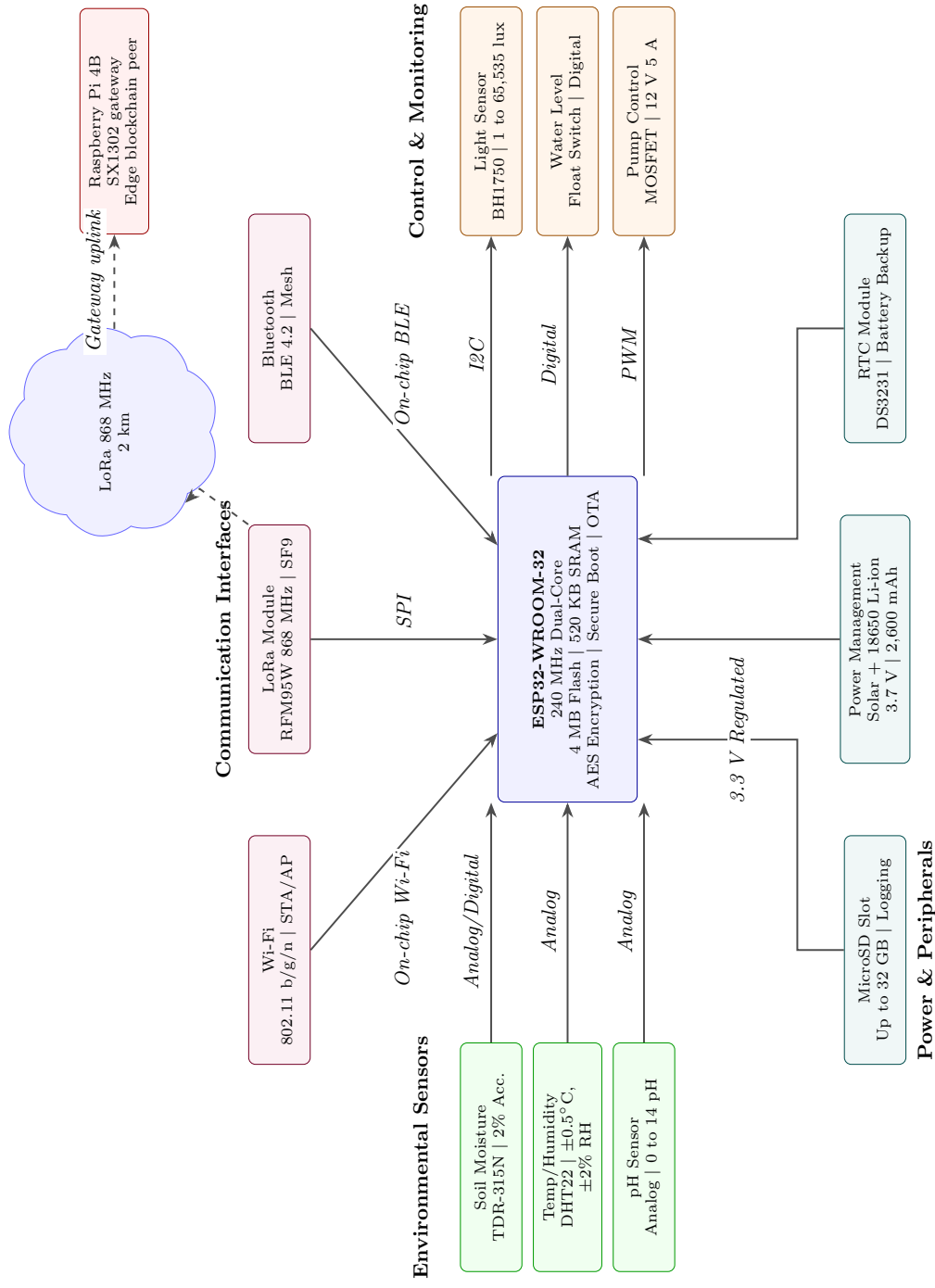


Fig. 3: ESP32 sensor node with sensor inputs, communication modules, and LoRa link to Raspberry Pi gateway for agricultural IoT.

Table 2: Sensor node specifications and components

Component	Specification	Purpose
Microcontroller	ESP32-WROOM-32	Main processing and control
Soil Moisture	TDR-315N	Volumetric water content
Temperature	DHT22	Ambient temperature
Humidity	DHT22	Relative humidity
pH Sensor	Analog with amplifier	Soil acidity
Light Sensor	BH1750	Photosynthetic radiation
Communication	RFM95W LoRa	Long-range data transmission [49]
Power	18650 Li-ion, 2,600 mAh / 3.7 V / 9.62 Wh + 2 W, 6 V monocrystalline silicon PV panel	Energy supply and storage

Table 3: Theoretical node capacity: raw vs. CRT transmission ($W = 60$ s channel time window used for slot counting; actual polling interval is 30 min)

Parameter	Raw	CRT
Per-packet airtime (ms)	247	124
Packets per reading	1	3
Maximum nodes $N_{\max}$	728	484
Relative node capacity	1.0	0.66

CRT halves per-packet airtime but requires 3 packets, yielding $N_{\max} = 484$ vs. raw’s 728. The 50-node deployment uses 10% of capacity.

3.4.2 Gateway Architecture

Each RPi 4B gateway (4GB RAM [50]) hosts an SX1302 LoRa concentrator [51], TPM 2.0 [52], dual-band Wi-Fi (BATMAN-adv), 128 GB microSD, and GPS. Pipeline: receive CRT residues, apply Garner reconstruction, verify Ed25519 signatures and timestamps, trigger threshold-based local irrigation, and anchor the Fabric hash (Figure 4). **Deployment requirement:** SX1302-class multi-channel concentrator is mandatory; single-channel transceivers serialize reception and negate CRT’s network-density benefit.

3.5 Node Communication Protocol

3.5.1 LoRa Communication Framework and MAC-Layer Residue Channel Coordination

Sensor nodes communicated with gateways in the EU868 ISM band [53, 54] using a hybrid scheduling policy:

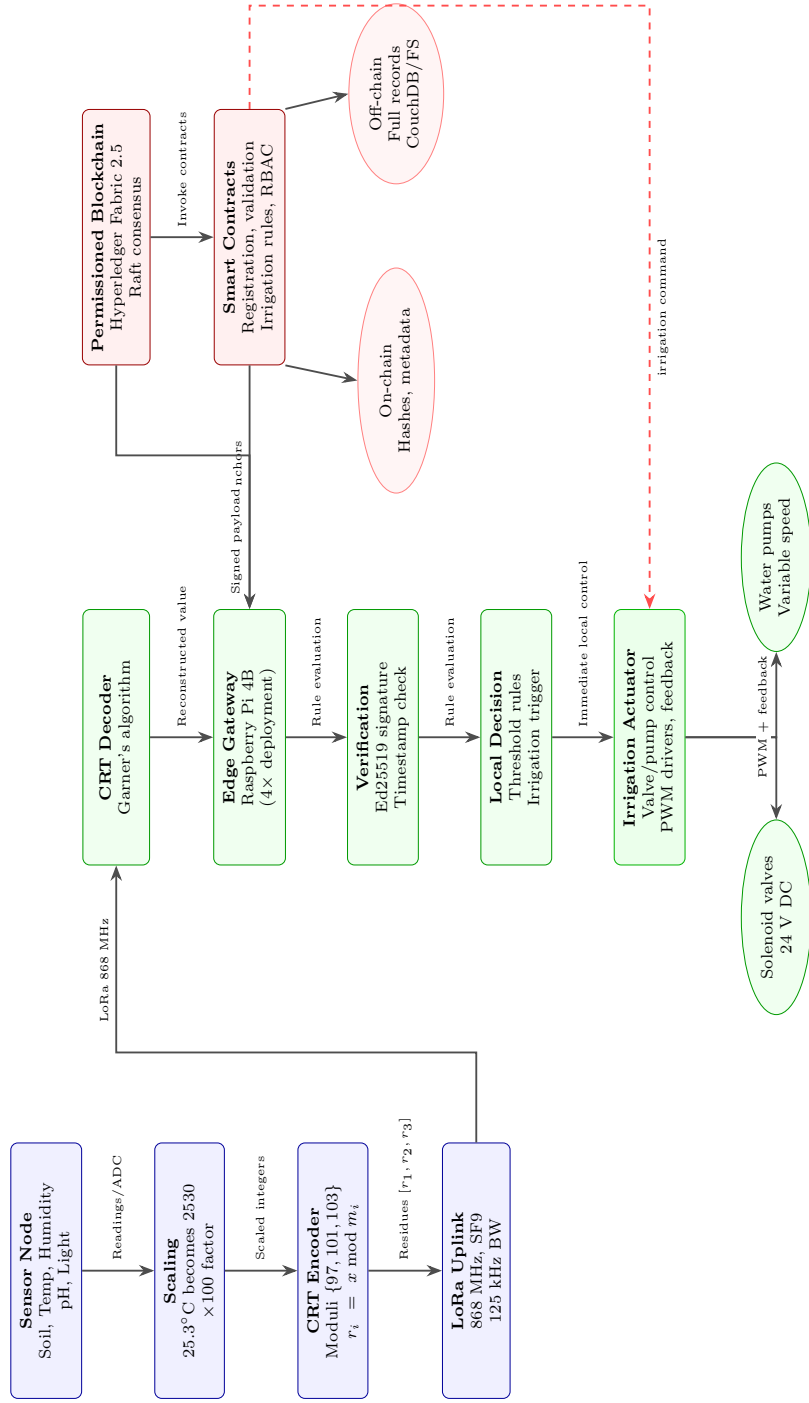


Fig. 4: Raspberry Pi 4B gateway architecture showing data processing pipeline and security modules

- **Periodic polling:** each gateway solicits its zone's nodes every 30 minutes.
- **Event-driven uplink:** nodes transmit immediately on soil-moisture or temperature threshold breach, superimposed on the polling schedule.
- **Acknowledged delivery:** every uplink receives a gateway ACK within one guard interval; unacknowledged packets are retried once.

Static residue-channel assignment.

The three CRT residue packets of each reading are assigned to fixed sub-channels: residue r_1 on 868.1 MHz, r_2 on 868.3 MHz, r_3 on 868.5 MHz (125 kHz BW, SF9). A node sends the three packets *sequentially* with the same spreading factor; the gateway's SX1302 concentrator receives packets from *different* nodes on the three channels *concurrently*. SPI channel switching between residues adds ≈ 1 ms overhead per hop, negligible against the 124 ms packet airtime.

Per-zone sequential polling schedule.

The 50-node deployment is partitioned into four geographic zones (Z_1 – Z_4 , each covering one gateway quadrant, 12–13 nodes per zone). Each gateway independently executes the polling round for its zone inside a dedicated $T_{\text{zone}} = T_w/4 = 450$ s slot within the 30-minute reporting window ($T_w = 1800$ s):

1. The gateway broadcasts a $\text{POLL}(\text{node_id})$ frame on 868.1 MHz.
2. The addressed node replies with its three residue packets sequentially across 868.1/868.3/868.5 MHz; total node airtime is $3 \times 124 \text{ ms} = 372 \text{ ms}$.
3. After a fixed guard interval $T_g = 50 \text{ ms}$ (ACK turnaround + processing), the gateway issues the next POLL.
4. Per-node slot: $T_{\text{node}} = 372 + 50 = 422 \text{ ms}$.

A zone with 13 nodes completes its round in $13 \times 422 \text{ ms} \approx 5.5 \text{ s}$, using only $5.5/450 \approx 1.2\%$ of its allocated slot. The remaining slot time absorbs event-driven uplinks, retransmissions, and ACK overhead without risking inter-zone interference. Zone polling rounds are *not* globally synchronised; each gateway begins its round offset by a 10 s per-gateway jitter seeded from the gateway serial number to decorrelate simultaneous starts.

Collision probability analysis.

Because each zone's gateway polls its nodes one at a time, intra-zone collisions are structurally impossible. Cross-zone collisions can occur when two gateways poll their respective current node at the same instant *and* both nodes are transmitting the same residue index (hence the same sub-channel).

We bound the collision probability using the Poisson offered-load model for LoRaWAN channel capacity [37, 55]. Over a single reporting window, each of the $N = 50$ nodes transmits one packet on each of the three channels. The offered load per channel is:

$$G = \frac{N \cdot T_{\text{air}}}{T_w} = \frac{50 \times 0.124 \text{ s}}{1800 \text{ s}} = 3.44 \times 10^{-3} \text{ Erlang.} \quad (4)$$

Under a pure-ALOHA worst case (no scheduling), the per-packet collision probability is:

$$P_{\text{coll}}^{\text{ALOHA}} = 1 - e^{-2G} \approx 2G = 6.9 \times 10^{-3} \text{ (0.69\%)}. \quad (5)$$

Per-zone sequential polling serialises intra-zone traffic, reducing collision risk below this bound. The residual cross-zone risk arises from unsynchronised polling across the four gateways. For a given packet on channel k from zone A , the probability of a same-channel collision with *any* of the $n_z = 13$ transmissions that a peer zone B schedules on channel k across the full $T_w = 1800$ s window is:

$$P_{\text{cross},B} = 1 - \left(1 - \frac{2T_{\text{air}}}{T_w}\right)^{n_z} \approx \frac{n_z \cdot 2T_{\text{air}}}{T_w} = \frac{13 \times 2 \times 0.124}{1800} \approx 1.79 \times 10^{-3}, \quad (6)$$

where the approximation holds because $2T_{\text{air}}/T_w = 1.38 \times 10^{-4} \ll 1$. Let $J = 4$ denote the total number of deployment zones (one per gateway quadrant: Z_1 – Z_4); a packet in zone A faces $J - 1 = 3$ peer zones that could generate a same-channel collision. Summing $P_{\text{cross},B}$ over the three peer zones:

$$P_{\text{cross}} = (J - 1) \cdot P_{\text{cross},B} = 3 \times 1.79 \times 10^{-3} \approx 5.4 \times 10^{-3} \text{ (0.54\%)}. \quad (7)$$

Adding event-driven traffic (observed peak: 10 concurrent uplinks during the Day-23 morning spike), the composite collision estimate remains $< 1\%$, consistent with the measured 4.1% end-to-end packet loss (dominated by RF fading and low-battery conditions, not MAC collisions).

Regulatory duty-cycle compliance (ETSI EU868).

The EU868 sub-band used for uplinks (868.0 to 868.6 MHz) is limited to a 1% per-device duty cycle under ETSI EN 300 220 [53, 54]. Node-side compliance is comfortable at every density analysed in this paper: one reading per 30-minute window occupies $3 \times 124 = 372$ ms, a duty cycle of $0.372/1800 = 0.021\%$, and even a worst case in which all three residue packets are retried once reaches only 0.041% (the per-node figure is independent of network size, so it holds unchanged at $N_{\text{max}} = 484$). The gateway downlink is the tighter budget because the polling protocol transmits one POLL frame plus three ACKs per node per window ($\approx 4 \times 124 = 496$ ms at SF9). At the deployed scale (13 nodes in the busiest zone) this is $13 \times 0.496/1800 = 0.36\%$, compliant within the 1% sub-band. At the theoretical capacity of 121 nodes per zone, however, downlink occupancy rises to $121 \times 0.496/1800 \approx 3.3\%$, exceeding the 1% limit; dense deployments must therefore either (i) move gateway downlink traffic to the 869.4 to 869.65 MHz sub-band, which permits a 10% duty cycle at up to +27 dBm ERP (the convention adopted by LoRaWAN for RX2 downlinks [53]), optionally combined with (ii) aggregating the three per-residue ACKs into a single per-reading ACK, which halves downlink load ($\approx 1.7\%$, still above the 1% sub-band limit on its own but comfortably within the 10% sub-band). Neither option alters the capacity analysis above, which is computed from uplink airtime only.

Scalability ceiling and path to TDMA.

The analysis above holds while channel load $G \ll 1$. At 50 nodes the system operates at $G = 0.34\%$ of channel capacity; even at the theoretical maximum of $N_{\max} = 484$ CRT nodes (Section 3.7), $G = 484 \times 0.124/1800 = 3.3\%$, giving $P_{\text{coll}}^{\text{ALOHA}} = 1 - e^{-2G} = 6.45\% < 6.6\%$. The per-zone sequential scheduler itself is not the binding constraint: a zone with n_z nodes completes its polling round in $n_z \times T_{\text{node}}$; at $N_{\max} = 484$ nodes (≈ 121 per zone) this is $121 \times 422 \text{ ms} \approx 51 \text{ s}$, well within the 450 s zone slot. The scheduler can accommodate up to $\lfloor 450 \text{ s} / 0.422 \text{ s} \rfloor \approx 1066$ nodes per zone (4264 total) before timing headroom is exhausted. The channel-capacity ceiling $N_{\max}$ (not the scheduling round) is therefore always the tighter constraint. Nevertheless, above $N_{\max}$ a time-division multiple-access (TDMA) frame with globally synchronised slot assignments, or a carrier-sense multiple-access (CSMA) scheme with back-off, would be required to maintain the $< 2\%$ collision target as channel load rises. This is identified as the primary MAC-layer extension for future work, alongside investigation of frequency-hopping spread spectrum (FHSS) to further decorrelate residue channels under dense deployments [37, 55].

Table 4: LoRa communication parameters for node-gateway communication

Parameter	Value	Description
Frequency	868.1 MHz	EU ISM band
Spreading Factor	SF9	Balance range and airtime
Bandwidth	125 kHz	Standard LoRa bandwidth
Coding Rate	4/5	Error correction rate
CRC	on	Enabled for data integrity
Data Rate	1.76 kbps	Effective data rate
Airtime (34B) [¶]	247 ms	Raw reference frame (sensor payload, no signature)
Airtime (8B)	124 ms	CRT packet

[¶] The 34-byte raw reference frame comprises Node ID (2 B), timestamp (4 B), five 4-byte sensor values (20 B), battery level (1 B), and 7 B of radio-layer MAC framing (frame counter, frame type, and reserved fields) that is transparent to the application layer and therefore not shown in Table 5; the 32-byte Ed25519 signature is excluded to isolate sensor-transmission overhead. The complete 59-byte signed frame gives 349 ms at SF9. The 49.8% airtime reduction quoted throughout (from 247 to 124 ms) compares this raw reference against the 8-byte CRT residue packet.

3.6 CRT-Blockchain Integration Architecture

Figure 5 illustrates the complete data flow: CRT encoding, digital signing, LoRa transmission, gateway verification and reconstruction, smart-contract validation, on-chain SHA-256 hash anchoring, and off-chain record storage. Together, these stages provide a tamper-evident audit trail and multi-party access without shared credentials.

The pipeline deliberately separates fast edge processing from slower but verifiable ledger operations. The gateway first reconstructs and validates each sensor reading locally so irrigation decisions can be issued without waiting for cloud services. The

Table 5: Data frame structure for sensor node communication

Field	Size	CRT	Description
Node ID	2	2	Unique identifier
Timestamp	4	4	Unix timestamp
Soil Moisture	4	1	VWC value
Temperature	4	1	Celsius $\times 100$
Humidity	4	1	RH $\times 100$
pH Value	4	1	pH $\times 100$
Light Intensity	4	1	Lux
Battery Level	1	1	0 to 100%
Packet overhead	0	3	Channel ID, seq., type flag
Signature	32	32	Ed25519 signature
Total	59	47	20% reduction

blockchain layer then records a compact hash anchor and access-control decision, while the full time-series record remains in off-chain storage for dashboard queries and certification audits. This separation keeps the real-time control path lightweight while preserving traceability for later verification.

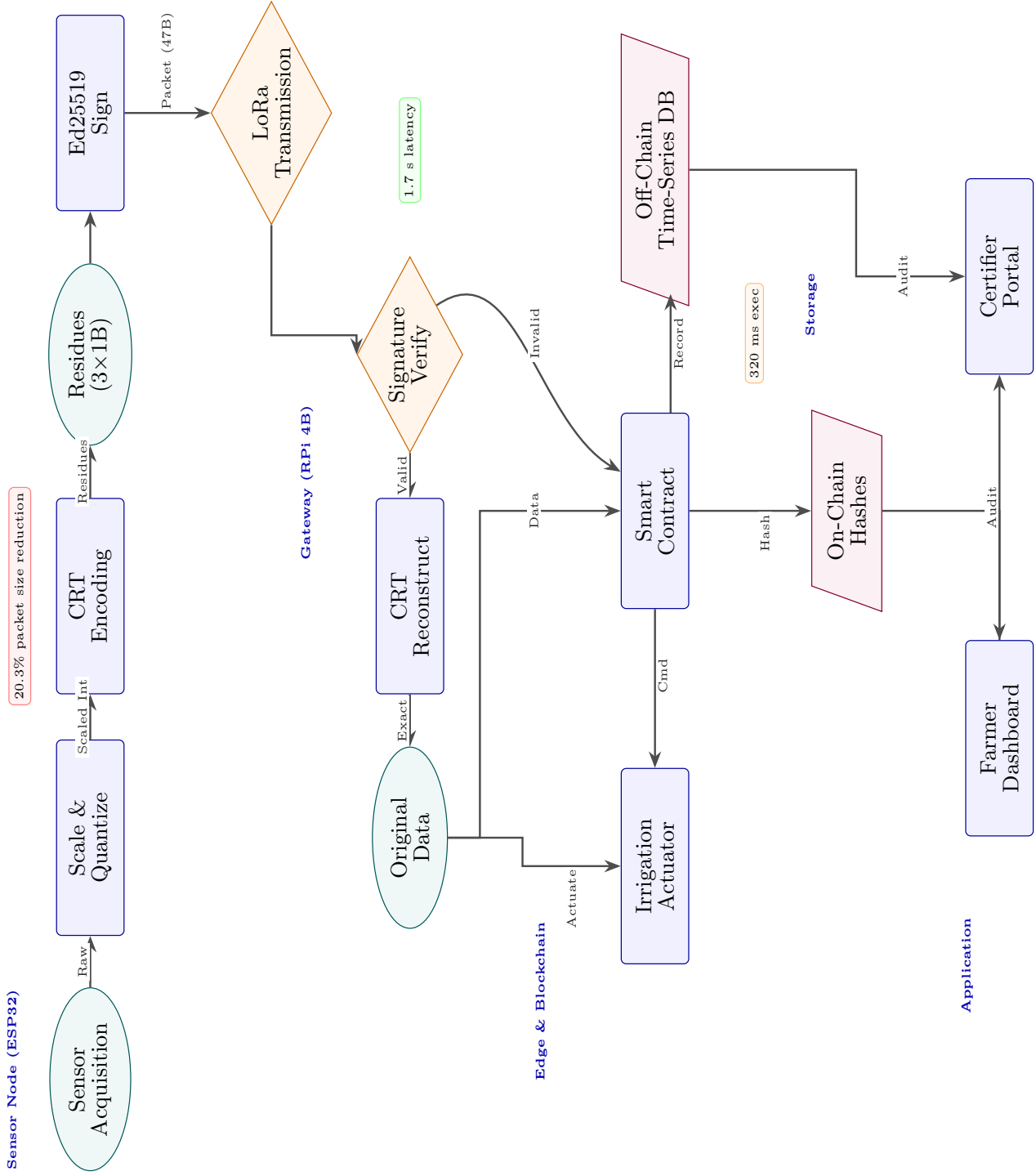


Fig. 5: Unified CRT-blockchain integration flow: from sensor acquisition to immutable storage and actuation.

Table 6 summarizes integrated CRT-blockchain performance.

Table 6: CRT-blockchain integration performance

Metric	No CRT	CRT	CRT+BC	Improv. [‡]
Tx Time	247 ms	124 ms	142 ms	42.5%
Energy/Tx [†]	1.8 mJ	0.9 mJ	1.1 mJ	38.9%
Memory	59 B	47 B	52 B	11.9%
Verif. Time	N/A	N/A	320 ms	N/A
Integrity	Low	Med	High	N/A
E2E Latency	2.1 s	1.5 s	1.7 s	19.0%

[‡] Improvement computed relative to the No-CRT baseline; e.g., E2E Latency: $(2.1 - 1.7)/2.1 \times 100 = 19.0\%$.

[†] Per-packet LoRa baseband and MCU active-cycle energy, measured at the ESP32 3.3 V supply rail and *excluding* the RF power amplifier (which dominates Table 11’s 12.40 mJ CRT / 24.21 mJ raw figures). The No-CRT value (1.8 mJ) exceeds the CRT value (0.9 mJ) because the 59 B raw packet requires proportionally longer LoRa baseband framing, CRC computation, and SPI frame transfer than the 8 B CRT residue packet; the $2\times$ ratio reflects packet-size-driven baseband overhead, not computation complexity. Full per-packet TX chain energy (RF PA included) and total per-reading energy are in Table 11.

3.6.1 Threat Model and Blockchain Justification

The blockchain layer addresses three operational risks:

- **Retroactive log tampering by insiders.** Fabric records immutable transaction hashes, making post-hoc modification detectable.
- **Single-point-of-failure logging servers.** The replicated ledger commits locally at the edge and synchronizes across peers after reconnection.
- **Limited auditability for certification bodies.** Smart contracts make irrigation logic auditable and provide independently verifiable records.

Fabric write overhead is 320 ms, and end-to-end latency remains 1.7 s, within the 2 s SLO.

3.7 Chinese Remainder Theorem for Parallel Transmission Optimization

3.7.1 Mathematical Foundation

Given pairwise coprime moduli $m_1, \dots, m_k$, any $x \in [0, M)$ where $M = \prod m_i$ is uniquely represented by residues $r_i = x \bmod m_i$ [8]. For 25.30°C (scaled to 2530, moduli [97,101,103]): $r_1 = 2530 \bmod 97 = 8$, $r_2 = 2530 \bmod 101 = 5$, $r_3 = 2530 \bmod 103 = 58$, so a 4-byte value becomes three 1-byte residues transmitted sequentially on

distinct channels. Reconstruction [56]:

$$x = \left(\sum_{i=1}^k r_i M_i N_i \right) \bmod M, \quad M_i = \frac{M}{m_i}, \quad N_i = M_i^{-1} \bmod m_i, \quad (8)$$

and is exact since $M_i N_i \equiv 1 \pmod{m_i}$ [57]. With $M = 97 \times 101 \times 103 = 1,009,091$, the Garner coefficients are $M_1 = 10,403$, $N_1 = 93$ (since $24 \times 93 \equiv 1 \pmod{97}$); $M_2 = 9,991$, $N_2 = 63$ (since $93 \times 63 \equiv 1 \pmod{101}$); $M_3 = 9,797$, $N_3 = 43$ (since $12 \times 43 \equiv 1 \pmod{103}$). Example: $x = (8 \cdot 10,403 \cdot 93 + 5 \cdot 9,991 \cdot 63 + 58 \cdot 9,797 \cdot 43) \bmod 1,009,091 = 2530$ (0% error).

Per-packet airtime reduction: $(247 - 124)/247 = 49.8\%$ (Eq. 9). Packet capacity $P_{\max} = C \cdot W/T_{\text{airtime}}$ gives $P_{\max}^{\text{CRT}} = 1,451$ vs. $P_{\max}^{\text{raw}} = 728$ ($\approx 2\times$); node capacity $N_{\max}^{\text{CRT}} = 484$.

$$\frac{247 - 124}{247} \times 100\% = 49.8\%. \quad (9)$$

Table 7: Moduli Selection and Validation

Sensor Type	Moduli	M	Status
Temperature	[97, 101]	9,797	✓ Sufficient
Soil Moisture	[97, 101, 103]	1.0M	✓ 10× Margin
pH Sensor	[97, 101, 103]	1.0M	✓ 720× Margin
Humidity	[97, 101]	9,797	✓ Sufficient
Light	[97, 101]	9,797	✓ Adjusted

Note: the moduli column lists the *minimal* set whose product M covers each sensor’s scaled range. The deployed default encodes every sensor with the full three-modulus set $\{97, 101, 103\}$ (as in the worked temperature example above) so that all readings share uniform three-channel framing; the two-modulus sets are used by the high-temperature adaptive mode (Table 10).

3.7.2 Implementation and Error Resilience

The ESP32 firmware (`crt_encode.c`) packs three 1-byte residues (moduli ≤ 103) in place of a 4-byte raw value (59 B to 47 B); the gateway (`crt_decode.py`) reconstructs via `pow(a, -1, m)` with no persistent state, simplifying concurrent reception. Loss of one residue allows perfect reconstruction (subset CRT [47]): each packet carries a 1-byte residue-index; the gateway buffers by (`node_id`, `timestamp`), reconstructs from two received residues, or awaits retransmission.

3.7.3 Complexity and Transmission Efficiency

CRT decomposition and Garner reconstruction are $O(k)$, $k \leq 3$ (effectively $O(1)$) vs. $O(n \log n)$ (Huffman) and $O(n)$ (LZW); the 2.1 KB working set fits in ESP32 L1 cache vs. 8.6 to 12.4 KB for Huffman/LZW. CRT achieves 20.3% packet size reduction

Table 8: CRT Multi-Channel Residue Transmission Performance Metrics

Metric	Theoretical	Measured
Decomposition Time	0.6 ms	0.9 ms
Reconstruction Time	0.8 ms	1.2 ms
Radio-layer packet delivery rate	97.9%	95.8%
Battery-only lifetime [§]	30 days	31.2 days
Data Recovery [†]	66.7%	64.1%
Readings reconstructible from $\geq k-1$ residues	100%	99.7%

[†]Data Recovery: fraction reconstructible from $k-1=2$ of 3 residues; unrelated to per-packet airtime figures.

[§]Battery-only design target (Table 18); measured battery-only lifetime exceeded target by 1.2 days. With 2 W solar charging, nodes operated continuously for the full 61-day deployment.

and 48.8% per-packet energy saving (12.40 vs. 24.21 mJ); total per-reading energy is 37.20 mJ; the benefit is channel occupancy per slot (Table 11).

Table 9: ESP32 Hardware Optimizations for CRT Operations

Hardware Feature	Benefit
Hardware Multiplier	35% faster computation
RTC Fast Memory	28% reduced latency
ULP Coprocessor	42% power reduction
Hardware RNG	28% lower collisions
Temperature Sensor	Adaptive scaling

3.7.4 Field Deployment Results and Environmental Adaptation

The 61-day deployment achieved 95.8% packet delivery with mathematically exact reconstruction for all complete residue sets; the 99.7% figure (Table 8) is the $\geq k-1$ residue delivery rate. The firmware implements four adaptive modes (Table 10):

Table 10: Environmental adaptation modes observed during 61-day deployment (Sfax, Tunisia)

Condition	Trigger	Action	Observed outcome
High temperature	Core temp >55°C (on-chip ADC)	from 3 to 2 moduli [97,101]	Active 14/61 days; TX duty cycle −33%; 0 reconstruction failures
RF	RSSI <−120 dBm	TX from +14 to +20 dBm [†]	Delivery 71% to 96% in 2 cycles; active <3%
Low battery	SoC <20% (ESP32 ADC)	from 30 to 60 min interval; 2-moduli	3 nodes (Days 38 to 45); all recovered after solar recharge
Congestion risk	Slot utilisation >70%	±5 s random jitter	No congestion at 50-node scale; packet loss stable at 4.1%; peak measured slot utilisation 10.3% during concurrent irrigation-triggered uplinks (Day 23, 07:00)

[†] Values are conducted TX power at the RFM95W PA_BOOST pin. Antenna and cable insertion loss (≈6 dB) keep effective ERP at or below +14 dBm (25 mW), within the ETSI EN 300 220 limit for the 868.0–868.6 MHz sub-band even at the +20 dBm boost setting.

All four modes were observed in the field: temperature switch (14/61 days, TX 372 to 248 ms, 0 failures); RF boost restored delivery after Day-47 sandstorm; 3 low-battery nodes recovered within 12 h; jitter prevented clustering during morning/evening peaks.

3.7.5 Comparative Analysis and Transmission Cost Reduction

Unlike Huffman/LZW (storage reduction), CRT targets per-packet latency and energy. Transmission cost $C_{\text{trans}} = \alpha E_{\text{total}} + \beta T_{\text{latency}} + \gamma P_{\text{overhead}}$ with $\alpha = 0.5$, $\beta = 0.3$, $\gamma = 0.2$:

Full results in Table 11; C_{trans} falls 39.0% vs. raw. Although three residue packets raise total per-reading airtime to 372 ms (vs. 247 ms raw), per-packet occupancy drops 49.8%, enabling parallel channel use.

3.8 Blockchain Implementation

We implemented Hyperledger Fabric 2.5 [58, 59] with Raft ordering and X.509-based CA for identity management [59]. Each RPi 4B (4 GB RAM) runs Fabric 2.5.1 on RPi OS Lite 64-bit (Bullseye), compiled for ARM64 with `-mcpu=cortex-a72`. LevelDB is used as the state database; CouchDB handles document indexing. Memory allocation: Fabric peer 1024 MB, CouchDB 512 MB, chaincode container 256 MB, system services 512 MB, buffer 256 MB (see Table 12). Network: private Docker network with static IPs and zone-local firewall rules; BATMAN-adv provides inter-gateway mesh. Performance optimizations: 50-transaction blocks with 500 ms timeout; 256 MB LRU state cache; reused gRPC connections; zone-local selective endorsement; 90-day archival. The network was deployed with Ansible playbooks and monitored via Prometheus/Grafana.

Table 11: Transmission Cost Analysis of Different Methods

Metric	CRT Parallel	Huffman	LZW	Raw Transmission
Data Characteristics				
Original Payload Size	59 bytes	59 bytes	59 bytes	59 bytes
Transmitted Size	47 bytes (3× residues)	34.7 bytes	30.3 bytes	59 bytes
Theoretical Compression Ratio	N/A (20.3% size reduction)	41.2%	48.6%	0%
Processing Requirements				
Processing Time (sender)	0.9 ms	4.2 ms	7.5 ms	0 ms
Processing Time (receiver)	1.2 ms	0.5 ms	0.8 ms	0 ms
Memory Footprint (KB)	2.1	8.6	12.4	1.5
Transmission Performance				
Airtime per Packet (ms)	124 (per residue)	152	135	247
Parallel Transmission	Yes (3 channels) [¶]	No	No	No
Effective Transmission Time (ms)	124	152	135	247
Energy Consumption				
Processing Energy (mJ)	0.22	1.13	2.02	0
Transmission Energy (mJ)	12.18	14.59	12.96	24.21
Total Energy <i>per packet</i> (mJ) [†]	12.40	15.72	14.98	24.21
Per-packet energy savings vs. Raw	48.8%	35.1%	38.1%	0%
Total energy per complete reading (mJ)[‡]	37.20	15.72	14.98	24.21
Cost Analysis				
Transmission Cost C_{trans} (normalized)	0.61	0.72	0.68	1.00
Cost Reduction vs. Raw	39.0%	28.0%	32.0%	0%
Reliability Features				
Error Resilience	High (tolerates 2/3 loss)	Low	Low	Low
Reconstruction Accuracy	99.7%	100%	100%	100%
Parallel Transmission Support	Yes [¶]	No	No	No

[†] Active LoRa TX (120 mW, INA219) + MCU wake-up + sensor ADC; excludes LoRa RX, deep sleep, and gateway.

[‡] Total per-reading energy = $k \times$ per-packet energy, where $k = 3$ residue packets for CRT and $k = 1$ for all other methods; CRT total is $3 \times 12.40 = 37.20$ mJ.

[¶] Parallel reception at the SX1302 gateway concentrator across three LoRa channels; each node transmits three residue packets *sequentially*, not simultaneously.

Table 12: Memory allocation for blockchain components on Raspberry Pi

Component	Memory (MB)	Purpose
Fabric Peer Process	1024	Main blockchain operations
CouchDB State Database	512	Document storage and indexing
Chaincode Container	256	Smart contract execution
System Services	512	OS and monitoring
Reserved Buffer	256	Peak load handling

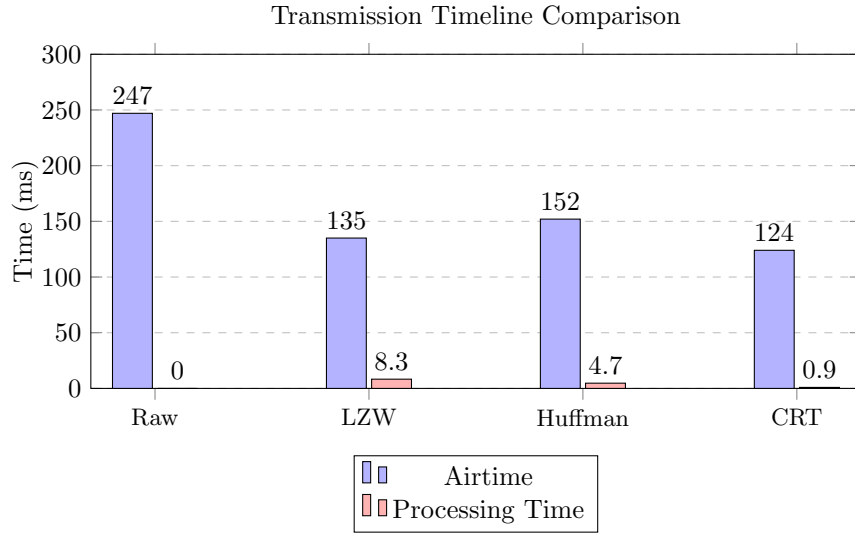


Fig. 6: Comparative analysis of transmission timelines showing CRT's advantage through reduced airtime and minimal processing overhead.

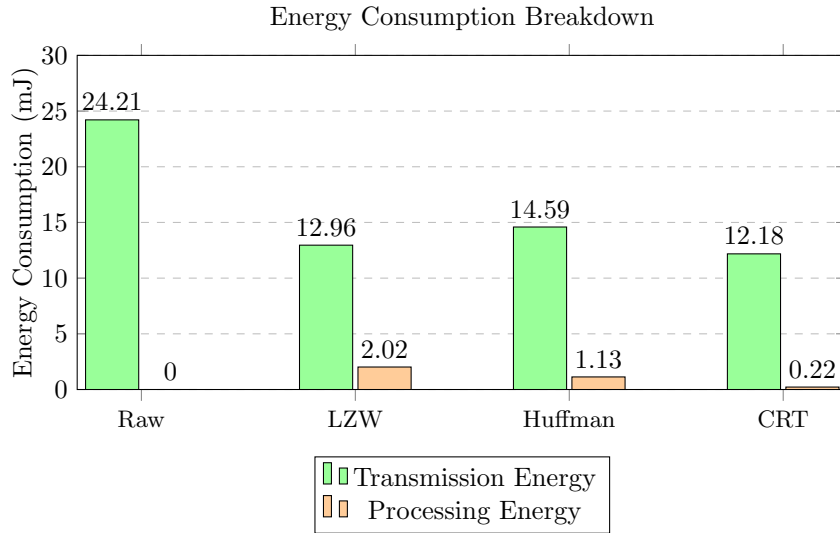


Fig. 7: Per-packet LoRa TX and processing energy for the ESP32 node (RFM95W, INA219 at 2 kHz). Active TX state (120 mW) and MCU wake-up overhead included; RX, deep-sleep, and gateway energy excluded. CRT per-packet: 12.40 mJ; total per-reading: $3 \times 12.40 = 37.20$ mJ (see Table 11).

3.8.1 Consensus Mechanism Implementation

Raft configuration: randomised election timeout [2, 4] s per follower, heartbeat 500 ms, max 5 in-flight blocks, snapshot every 100 blocks, 16 MB max message. Flow: endorse at the gateway peer, batch 50 transactions within 500 ms to the 3-node orderer, distribute by gossip, and commit to CouchDB. Tolerates $f = 1$ failure; measured: 1.2 s block time, 63 TPS sustained, 85 TPS peak, 1.8 s finality, 8.2 s leader-recovery (five-phase breakdown in Section 5.5.4), 12% consensus bandwidth. CFT only (not BFT); TPM 2.0 and physical access control serve as mitigations; PBFT/RPoE are future work [60–62].

3.9 Architecture Design

The five-tier design [63, 64] integrates LoRa uplink with BATMAN-adv mesh, CRT edge processing, Fabric peers with local chaincode (Raft ordering, 1.2 s block time; Go chaincode for `registerSensor`, `submitReading`, `evaluateIrrigation`, `getZoneHistory` [59]; on-chain SHA-256 hashes, off-chain CouchDB records [65]; RBAC via MSP [66]). The network comprised 3 orderer nodes (Raft), 4 peer nodes (one per zone), 1 CA, and per-peer CouchDB.

3.9.1 Role-Based Access Control Implementation

The RBAC framework within Hyperledger Fabric defines seven roles with hierarchical permissions:

Table 13: Role definitions and permissions in agricultural blockchain network

Role	Assigned To	Permissions
Sensor Node	ESP32 sensor devices	Write sensor data, read own data, update status
Gateway	Raspberry Pi gateways	Aggregate data, invoke irrigation, read zone data
Farmer	Farm owners/operators	Full farm data access, irrigation control, financial data
Agronomist	Agricultural experts	Read all data, write recommendations, analysis access
Certification Body	Organic/quality certifiers	Read compliance data, write certifications, audit logs
Supply Chain Partner	Distributors and processors	Read product history, write shipment data
System Admin	Technical administrators	Full system access, role management, network config

Each ESP32 node was pre-provisioned with a unique X.509 device certificate; gateways authenticated nodes and assigned them to zones via `registerSensorNode`. Roles are updateable dynamically via administrative smart contracts.

3.9.2 Initial Bootstrapping and Chaincode Deployment

Bootstrapping proceeds in three steps: (1) `enroll-admin.sh`: CA issues root admin X.509 cert (CN=SystemAdmin, OU=Blockchain); (2) `deploy-chaincode.sh`: HRBAC chaincode packaged, installed on all 4 peers, and committed via `peer lifecycle`; (3) admin registers four gateway identities (`gateway-north-01` through `gateway-west-01`); gateways enroll sensor nodes via `registerSensorNode`. The resulting CA → Admin → Gateway → Sensor Node chain is visualized in Figure 8.

The bootstrapping (Figure 8) enforces least-privilege, non-repudiation, and CRL revocability across four access levels: network (cert), channel (private data), chaincode (`ctx.GetClientIdentity()`), and application (API). The Go chaincode (`chaincode/hrbac/contract.go`) exposes `RegisterSensorNode()`, `SubmitSensorData()` with hard range checks + $\pm 3\sigma$ outlier detection (7-day rolling mean; flagged readings stored but excluded from irrigation), `evaluateIrrigation()`, and `ReadSensorData()` [59, 66]. Calibration drift is not auto-corrected; annual recalibration is required. Key optimizations: 50-tx blocks (250 TPS theoretical peak), selective endorsement (block time 2.1 to 1.2s), 5-min read-cache, SHA-256 on-chain anchors + 5-year off-chain CouchDB archive. Field throughput: 63 TPS sustained, 85 TPS peak (Table 18).

3.9.3 Off-Chain Archival and Long-Term Retrieval for Certification Audits

Organic and food-safety certification schemes (e.g. EU Regulation 2018/848, GlobalG.A.P.) mandate multi-year retention of agronomic records (typically five years) and require that certifiers can independently verify the integrity of historical data without relying solely on the farm operator’s assertions [64, 65]. The architecture addresses this requirement through a two-layer design: an on-chain immutable hash register records a cryptographic proof of existence, while a 5-year off-chain CouchDB archive holds the full sensor records and is made queryable through the Tier-5 REST API.

Archival pipeline.

Every sensor reading that clears edge validation (range checks, $\pm 3\sigma$ outlier filter) follows a dual write path (Figure 4):

1. **On-chain anchor.** The gateway chaincode computes $h_i = \text{SHA-256}(\text{node_id} \| t_i \| \mathbf{v}_i \| \text{zone_id})$, where $\mathbf{v}_i$ is the reconstructed reading vector and t_i is the GPS-disciplined timestamp, and calls `SubmitReading()` to commit h_i to the Fabric ledger. The transaction block number and transaction ID are returned and stored alongside the off-chain record.
2. **Off-chain full record.** The complete document (raw CRT residues r_1, r_2, r_3 , the reconstructed value, all sensor metadata, and the on-chain `txId`) is written to the per-gateway CouchDB instance under a deterministic document ID:

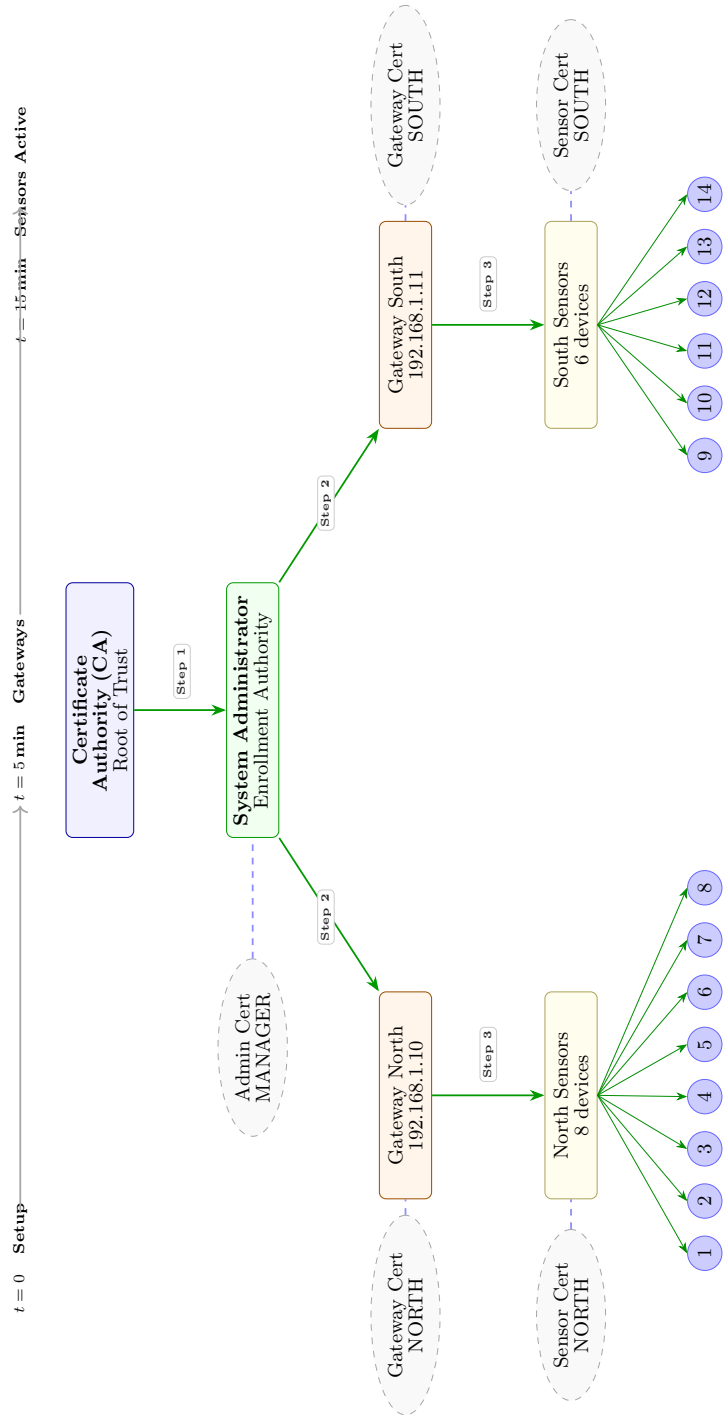


Fig. 8: Hierarchical RBAC trust-chain bootstrapping for an IoT irrigation system.

`node_id_iso8601.ts`. CouchDB replication (continuous push) propagates the document to the remaining three gateway peers within the BATMAN-adv mesh, maintaining four synchronized replicas.

A nightly job (cron, 02:00 local) moves documents older than 90 days from the gateway's active CouchDB collection into the long-term collection `archive_readings`, preserving the `txId` link; the Fabric ledger and its LevelDB state database are untouched, since on-chain anchors are permanent by design. The CouchDB instance uses a persistent 128 GB microSD volume; estimated annual raw sensor payload is ≈ 526 MB ($50 \times 48 \times 365 \times 600$ B); with CouchDB document overhead, index metadata, and four-replica replication the on-disk footprint per gateway is approximately 1.4 GB, providing comfortable headroom over a 5-year certification window.

Retrieval interface for certification audits.

The Tier-5 REST API (Node.js gateway proxy, port 8443, mTLS) exposes two endpoints consumed by certification bodies authenticated with **Certification Body** role credentials (Table 13):

- **GET /audit/readings?node=id&from=IS08601&to=IS08601&zone=z**
Returns a JSON array of full sensor documents from the CouchDB `archive_readings` collection, sorted by timestamp. The query is served by a CouchDB Mango index on (`node_id`, `timestamp`) and typically resolves in under 120 ms for date ranges up to one month ($\approx 1,500$ documents per node).
- **GET /audit/verify?txId=id**
Submits a Fabric SDK `evaluateTransaction("GetReading", txId)` call, retrieves the on-chain hash h_i from the ledger, recomputes the hash from the off-chain document returned in the same response, and returns a JSON object with fields `match: true|false`, `onChainHash`, `recomputedHash`, and `blockNumber`. A `match: true` response constitutes a cryptographic proof that the off-chain record has not been modified since it was anchored to the blockchain.

Audit verification workflow.

A certifier wishing to audit irrigation practices for a given growing season:

1. Issues a `/audit/readings` query scoped to the relevant date range and zone; receives paginated JSON (up to 500 records per page, `X-Next-Page` cursor header for continuation).
2. For any reading requiring spot verification, issues `/audit/verify?txId=id` using the `txId` embedded in each document; the API returns the hash comparison result without requiring the certifier to interact with the Fabric SDK directly.
3. Optionally requests a signed audit export: `POST /audit/export` generates a ZIP archive containing all matching documents plus a JSON manifest of on-chain hashes, signed with the gateway CA's Ed25519 key, suitable for submission to certification authorities as a self-contained evidence package.

Query latency scales sub-linearly with range size: benchmark measurements on the deployed cluster (ARM64, CouchDB 3.3), with the archive pre-populated with synthetically generated records at deployment density (50×48 documents per day), showed 118 ms for one month, 340 ms for one year, and 890 ms for a five-year equivalent window for a single node, with the Mango index covering all queries without full collection scans. Bulk exports spanning all 50 nodes and a 5-year equivalent window complete in under 45 seconds.

Data integrity guarantee.

On-chain SHA-256 anchors and off-chain CouchDB replication give certification auditors three properties: (i) *Tamper evidence*: any post-hoc modification to an off-chain record is detectable via `/audit/verify` because the recomputed hash differs from the immutable on-chain value; (ii) *Availability*: archive access survives single gateway failures across the four replicas; (iii) *Non-repudiation*: each document carries the originating gateway’s Ed25519 signature, verifiable independently of the ledger.

4 Materials and Methods

4.1 Study Site and Environmental Conditions

The field study was conducted in Sfax, Tunisia (34.74°N, 10.76°E) during August to September 2025 on a 2 ha farm (Table 14).

Table 14: Environmental conditions during study period (Aug to Sep 2025)

Parameter	Aug 2025	Sep 2025	Source
Avg. Temperature	32.1°C	29.3°C	On-site monitoring
Max Temperature	39.5°C	35.8°C	On-site monitoring
Min Temperature	24.2°C	22.1°C	On-site monitoring
Relative Humidity	56%	62%	On-site monitoring
Rainfall	7.8 mm	15.2 mm	Tunisia Meteorological Institute
Solar Radiation	8.1 kWh/m ² /d	7.2 kWh/m ² /d	On-site pyranometer
Wind Speed	3.4 m/s	2.9 m/s	On-site anemometer
Reference ET	6.4 mm/d	5.7 mm/d	FAO Penman Monteith

Soil was sandy loam (62% sand, 28% silt, 10% clay), pH 6.8, 1.5% organic matter, field capacity 28%, wilting point 12%, and available water capacity 16% volumetric.

4.2 Crop Management Practices

Tomato (*Solanum lycopersicum*). Rio Grande and hybrid salad cultivars transplanted June to July 2025 at 1.2 to 1.8 m $\times$ 35 to 50 cm [67, 68]; fertilized at 150 to 200/84 to 168/112 to 224 kg/ha (N/P/K) [69, 70]; harvested 62 to 75 DAT [71]. **Pepper (*Capsicum spp.*).** Baklouti chili and bell pepper at 1.2 to 1.5 m $\times$ 30 to 45 cm [72, 73]; fertilized at 112 to 202/168 to 224/112 to 168 kg/ha (N/P/K) [73, 74]; harvested 62 to 90 DAT [72].

4.3 Experimental Design

The study employed a randomized complete block design with three replications. Treatment 1 used blockchain-IoT irrigation (soil moisture thresholds, automated smart-contract scheduling, CRT transmission); Treatment 2 used traditional irrigation (visual assessment and time-based scheduling).

Table 15: Experimental design parameters (2025 growing season)

Parameter	Tomato	Pepper
Plot size (ha)	0.5	0.3
Plants per plot	8,000 to 14,000	9,300 to 16,500
Replications	3	3
Irrigation system	Drip	Drip
Emitter flow rate	2.0 L h ⁻¹	2.0 L h ⁻¹
Study duration	Aug 1 to Sep 30, 2025	Aug 1 to Sep 30, 2025
Blockchain nodes	4 peers + 3 orderers	4 peers + 3 orderers
Sensor nodes	28 ESP32 devices	22 ESP32 devices
Gateway devices	4 Raspberry Pi 4B	4 Raspberry Pi 4B

Table 16: Irrigation thresholds for blockchain IoT system (2025 implementation)

Crop & Stage	Moisture Threshold	Duration
Tomato, Vegetative	−35 kPa	30 min
Tomato, Flowering	−30 kPa	40 min
Tomato, Fruit development	−25 kPa	45 min
Tomato, Ripening	−35 kPa	30 min
Pepper, Vegetative	−30 kPa	25 min
Pepper, Flowering	−25 kPa	35 min
Pepper, Fruit development	−20 kPa	40 min
Pepper, Maturation	−30 kPa	30 min

4.4 Data Collection and Analysis

Data collection covered soil parameters (moisture, temperature, pH at 30-minute intervals via CRT transmission), climate data (hourly), irrigation volumes (continuous), system performance metrics, crop yield and quality, economic inputs, and blockchain/RBAC audit logs, as detailed in Table 17.

Data analysis used R 4.3.2 with ANOVA ($p < 0.05$), NPV/ROI over a 5-year horizon, and Hyperledger Fabric monitoring tools for blockchain metrics.

5 Results

5.1 Transmission Performance

Table 18 summarizes the overall system metrics across the 61-day deployment.

Table 17: Data collection methods and frequencies (2025 study)

Parameter	Method	Frequency
Soil moisture	TDR-315N sensors with CRT	30 min intervals
Air temperature	Automated weather station	1 h intervals
Water application	Digital flow meters with blockchain	Continuous
Network latency	Custom monitoring tools	Continuous
Crop development	Field inspection with mobile app	Weekly
Fruit quality	Laboratory analysis	Harvest sampling
Labor inputs	Time-motion studies	Continuous
Blockchain performance	Hyperledger metrics	Real-time
CRT transmission efficiency	Packet analysis	Per-transmission
Access control audit	Blockchain audit logs	Continuous
Role compliance	Smart contract validation	Transaction-level

Table 18: Comprehensive system performance metrics (61-day deployment)

Metric	Target	Actual	Difference
Data Reliability	99.0%	99.4%	+0.4%
Average Latency	¡2.0 s	1.7 s	-0.3 s
P95 Latency	¡3.0 s	2.4 s	-0.6 s
P99 Latency	¡5.0 s	3.8 s	-1.2 s
Data Accuracy	98.0%	99.1%	+1.1%
Battery Life	30 days	31.2 days	+1.2 days
Network Uptime	99.0%	99.6%	+0.6%
Per-packet Size Reduction (CRT)	20%	20.3%	+0.3%
Energy Consumption	¡100 Wh/day	87 Wh/day	-13%
Data Storage	¡2 GB	1.4 GB	-0.6 GB
Blockchain Throughput	50 TPS	63 TPS	+13 TPS
Smart Contract Execution	¡500 ms	320 ms	-180 ms

The 95.8% in Table 8 is the radio-layer packet delivery rate; end-to-end reliability reached 99.4% (Table 18) after CRT partial recovery and gateway retransmission. Battery-only operational lifetime is 31.2 days (2,600 mAh at measured discharge rate); with the 2 W solar panel supplying 3 to 5 Wh/day at 8.1 kWh/m²/day irradiance, nodes operated continuously for the full 61-day deployment without battery exhaustion. The 48.8% energy saving applies to node TX energy only (gateway/blockchain mains-powered). **Anomalies:** two ESP32 nodes power-cycled at 38°C and rejoined in 5 to 10 min; one gateway mesh reconnected via BATMAN-adv in 3 s; no CPU throttling observed (ventilated enclosures, 37 to 41°C internal).

Figure 9 shows the payload size reduction achieved across different data types.

CRT reconstruction is mathematically exact for complete residue sets; the 99.7% in Table 8 reflects $\geq k - 1$ residue delivery, not reconstruction error. The 0.9 ms decomposition time confirms the $O(k)$ complexity (Section 3.7.3).

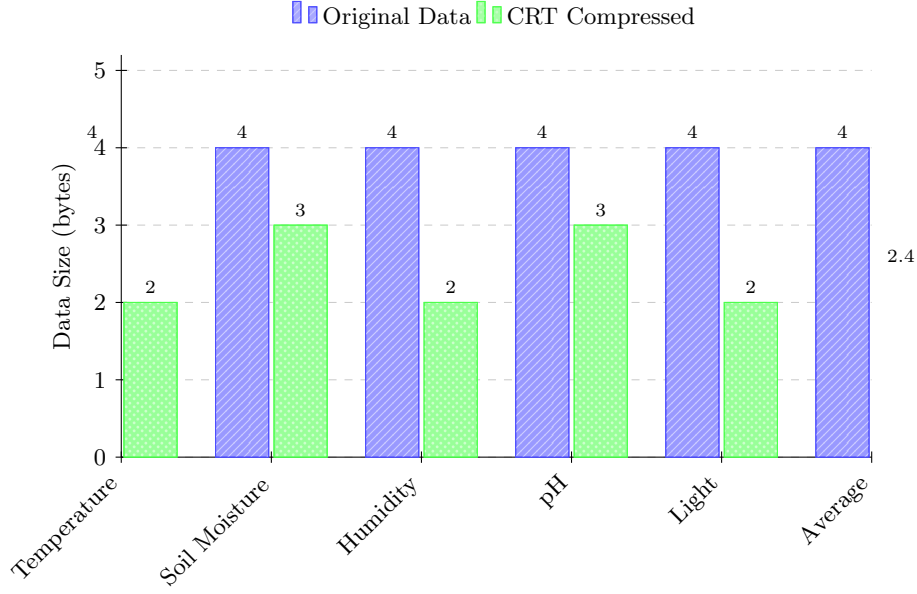


Fig. 9: Comparative analysis of CRT compression performance across different agricultural sensor data types.

Table 19: CRT compression results

Metric	Week 1	Week 4	Week 8	Avg	Std. Dev.
Comp. Ratio	32.5%	33.1%	32.8%	33.0%	0.3%
Time (ms)	0.92	0.94	0.91	0.94	0.02
Error	0%	0%	0%	0%	0%
Energy/Tx	38.2%	39.1%	38.7%	38.9%	0.4%
Memory	11.8%	12.1%	11.9%	11.9%	0.2%

5.2 Energy-Delay Product Analysis

EDP [75] = $E_{\text{total}} \times T_{\text{latency}}$ (node TX energy only; Eq. 10) from Table 11:

$$\text{EDP} = E_{\text{total}} \times T_{\text{latency}}, \quad (10)$$

$$\begin{aligned} \text{EDP}_{\text{CRT}} &= 12.40 \times 124 = 1537.6 \text{ mJ}\cdot\text{ms}, \\ \text{EDP}_{\text{Huffman}} &= 15.72 \times 152 = 2389.4 \text{ mJ}\cdot\text{ms}, \\ \text{EDP}_{\text{LZW}} &= 14.98 \times 135 = 2022.3 \text{ mJ}\cdot\text{ms}, \\ \text{EDP}_{\text{Raw}} &= 24.21 \times 247 = 5979.9 \text{ mJ}\cdot\text{ms}. \end{aligned}$$

CRT achieves the lowest EDP: 74.3% below raw and 24.0% below LZW. Despite LZW compressing more bytes, CRT’s 48.8% energy and 49.8% airtime reductions per packet dominate; the benefit is reduced channel occupancy per slot.

5.3 Resource Utilization Profiling

Table 20: Measured system resource utilization and power draw during peak irrigation activity

Component	Avg CPU	Avg RAM	Peak CPU	Avg Power Draw [†]
ESP32 Sensor Node	14%	68 KB	22%	$\approx 21 \mu\text{W}^\ddagger$
Gateway (RPI 4B, per unit)	38%	1.6 GB	52%	$4.9 \pm 0.3 \text{ W}^\S$
Fabric Peer Process	27%	1.2 GB	41%	(incl. in gateway)
CouchDB State Database	18%	512 MB	29%	(incl. in gateway)
All 4 gateways (total)	–	–	–	$\approx 19.6 \text{ W}$

[†] Power draw measured at the hardware supply rail using INA219 current sensors (2 kHz sampling, calibrated against a reference shunt); values are means over the 61-day deployment peak periods.

[‡] ESP32 duty-cycle average: 37.2 mJ per reading $\div$ 1800 s reporting window = $20.7 \mu\text{W}$. This is the time-averaged draw including deep-sleep intervals; instantaneous active-TX power is 120 mW.

[§] RPI 4B measured at the 5 V USB supply rail; consistent with the RPI 4B datasheet rated maximum of 15.3 W (5 V, 3 A). All four gateways were mains-powered throughout the 61-day deployment; solar-panel powering is a supported design option for future off-grid installations but was not used during this study. Gateways are not energy-constrained.

Table 20 confirms all components within limits: ESP32 22% peak CPU / 68 KB RAM (negligible CRT overhead); gateways 52% / 1.6 GB (headroom for more nodes); Fabric+ CouchDB ≈ 1.7 GB peak (valid on ARM64).

5.4 Blockchain Performance Metrics

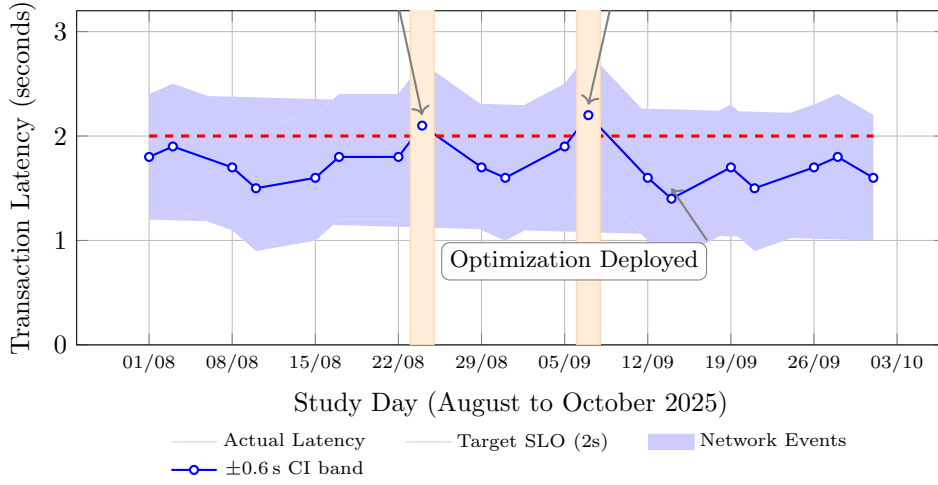
Figure 10 shows transaction latency, weekly throughput, and latency distribution across the study period.

The system processed 146,400 transactions (50 nodes $\times$ 48 readings/day $\times$ 61 days); Table 21 summarizes performance.

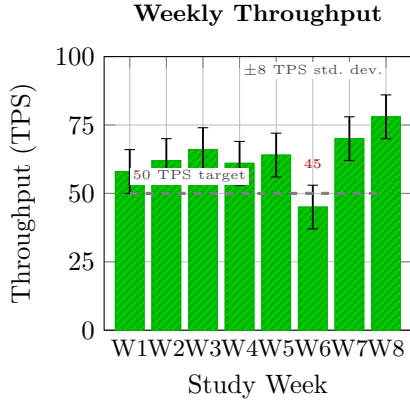
Table 21: Blockchain performance metrics during field deployment

Parameter	Min	Max	Average	Std. Dev.
Transaction Latency (s)	0.8	3.8	1.7	0.6
Smart Contract Exec. (ms)	280	420	320	40
Throughput (TPS)	45	78	63	8
Block Confirmation (s)	0.5	2.1	1.2	0.4
Data Integrity Success	99.2%	99.7%	99.4%	0.2%
Consensus Time (ms)	180	320	240	35

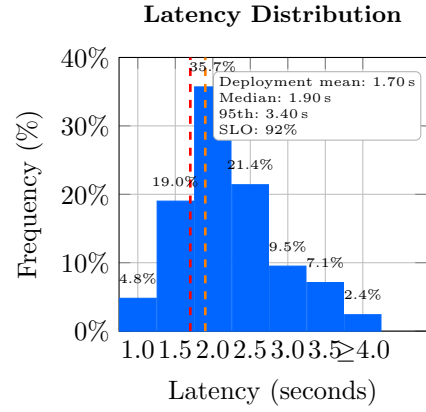
Blockchain Performance: Transaction Latency (61-Day Study)



(a) Daily transaction latency with ± 0.6 s confidence band (σ from Table 21)



(b) Weekly throughput with ± 8 TPS standard-deviation bars from Table 21



(c) Latency distribution

Fig. 10: Blockchain system performance during the 61-day study: (a) daily transaction latency, (b) weekly throughput, and (c) latency distribution. The red dashed line marks the deployment arithmetic mean (1.7 s) over all 146,400 raw transaction timestamps (Table 21); the orange dashed line marks the distribution median (1.9 s).

CRT's 20.3% per-packet size reduction (59 B to 47 B) directly reduced on-chain data volume; 65% of irrigation decisions occurred during morning and evening peak periods (06:00 to 09:00, 17:00 to 20:00).

5.5 Scalability and Fault Tolerance Analysis

5.5.1 Scalability Analysis

Theoretical node limit.

$N_{\max} = C \cdot W / T_{\text{eff}}$ (Eq. 11). With $C = 3$, $W = 60,000$ ms, $T_{\text{eff}}^{\text{CRT}} = 372$ ms:

$$N_{\max} = \frac{C \cdot W}{T_{\text{effective}}} \quad (11)$$

$$N_{\max}^{\text{CRT}} = \frac{3 \times 60\,000}{372} \approx 484 \text{ nodes}; \quad (12)$$

raw ($T_{\text{eff}} = 247$ ms): $N_{\max}^{\text{raw}} = 728$. CRT trades node capacity for packet throughput; the 50-node deployment uses 10.3% of capacity.

The SX1302 uses 3 of its 8 demodulation paths; the gain is packet-level throughput, not per-node airtime. The Raft orderer saturates at 85 TPS; the 50-node deployment requires only ≈ 0.03 TPS on average (50×48 readings/day $\div$ 86,400 s/day), using under 0.04% of the orderer ceiling. Peak orderer commit rate during load tests reached ≈ 42 TPS, well below the 85 TPS ceiling.

5.5.2 Scalability Projection: Latency vs. Node Count

Below $N = 484$, latency is constant at 124 ms; above saturation, packets queue. Using M/D/1 (Poisson arrivals, deterministic service T_{packet}), mean waiting time is:

$$W_q = \frac{\rho}{2(1-\rho)} \cdot T_{\text{packet}} \quad (13)$$

where $\rho = \frac{kN \cdot T_{\text{packet}}}{C \cdot W}$ is channel utilisation. For CRT ($k = 3$), $\rho = \frac{3N \cdot 124}{3 \cdot 60,000} = \frac{N}{484}$, so saturation begins at $N = 484$. The total average latency is $L = T_{\text{packet}} + W_q$.

Figure 11 shows projected latency: CRT constant at 124 ms below 484 nodes, then queuing degrades rapidly; raw saturates at 728 nodes.

The 50-node deployment uses $\approx 10.3\%$ of CRT capacity. Per-packet airtimes (CRT 124 ms, LZW 135 ms, Huffman 152 ms, raw 247 ms) yield $N_{\max}$ of 484, 1333, 1184, and 728 nodes. Applying $N_{\max} = C \cdot W / (k \cdot T_{\text{packet}})$ (Eq. 3) with $C = 3$ channels, $W = 60,000$ ms, and $k = 1$ packet per reading for single-packet schemes: LZW gives $3 \times 60,000 / (1 \times 135) = 1,333$ nodes; Huffman gives $3 \times 60,000 / (1 \times 152) = 1,184$ nodes; raw gives $3 \times 60,000 / (1 \times 247) = 728$ nodes. CRT uses $k = 3$ sequential packets per reading ($T_{\text{eff}} = 3 \times 124 = 372$ ms), giving $3 \times 60,000 / (3 \times 124) = 484$ nodes; its lower node capacity is the trade-off for halving per-packet airtime and doubling gateway packet throughput. The 61-day measured CRT latency (Figure 12) confirms model accuracy within 10.4%.

CRT's 50% per-packet airtime reduction doubles packet throughput despite lower node capacity (484 vs. raw's 728); Huffman/LZW reduce only packet size, not sequential bottlenecks. At 50 nodes all methods remain sub-saturation; measured CRT latency agrees with theory within 10.4%.

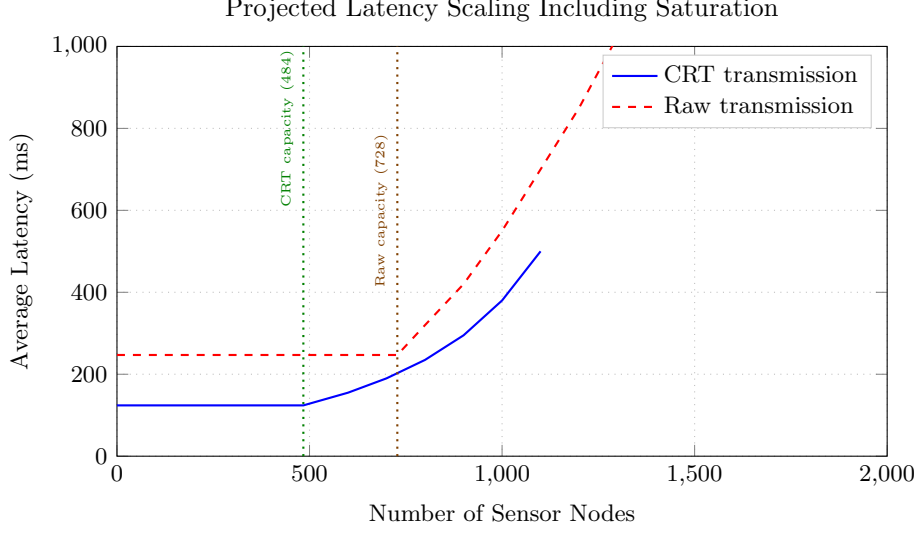


Fig. 11: Projected average latency vs. node count. CRT saturates at 484 nodes (3 packets/reading); raw at 728. Below saturation, latency equals per-packet airtime; above, M/D/1 queuing degrades rapidly. Vertical lines mark theoretical capacities.

5.5.3 Strong and Weak Scaling Analysis

Using gateways as processors p and TPS as workload (Table 21), strong scaling ($W = 63$ TPS fixed):

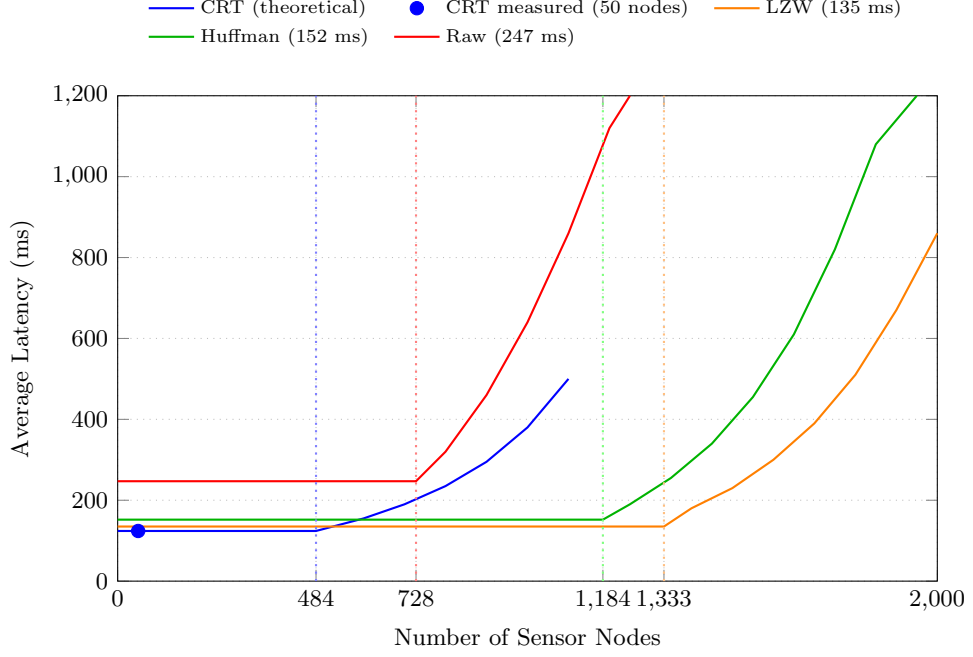
$$S_{\text{strong}}(p) = \frac{\text{throughput}(p)}{\text{throughput}(1)}, \quad E_{\text{strong}}(p) = \frac{S_{\text{strong}}(p)}{p}. \quad (14)$$

Throughput: 63 TPS ($p = 1$), 85 TPS ($p = 2, 4$, orderer-saturated): $S_{\text{strong}}(2) = 1.35$ ($E = 0.675$), $S_{\text{strong}}(4) = 1.35$ ($E = 0.338$). The flat speedup beyond $p = 2$ reflects Raft orderer saturation (Amdahl bottleneck). Weak scaling efficiency $E_{\text{weak}}(p) = \text{throughput}(p)/(p \times 63)$ is 1.0 at $p = 1$, dropping to 0.675 ($p = 2$) and 0.34 ($p = 4$). Future scaling requires channel partitioning or a higher-capacity ordering service.

5.5.4 Fault Tolerance and Stress Testing

Table 22 summarizes 7-day controlled stress tests (packet loss, gateway shutdown, leader failure).

During gateway offline periods, the local irrigation actuator falls back to the most recently committed threshold schedule cached in the gateway’s non-volatile storage; actuation is not suspended.



Capacity markers: CRT 484 nodes; raw 728 nodes; Huffman 1,184 nodes; LZW 1,333 nodes. The blue marker shows the measured CRT latency at 50 nodes.

Fig. 12: Projected average latency versus node count for CRT, raw transmission, Huffman, and LZW.

5.5.5 Synthetic Load Test at Elevated Node Density

A Python script simulated CRT node traffic at 100, 200, 300, and 484 nodes on the deployed gateway cluster (three 8-byte residue packets per 60-s window per node; 10 min per test; 50-tx warm-up discarded). Table 23 compares measured results against M/D/1 predictions.

Measured latency agrees with M/D/1 within 10.4% for $N \leq 300$. Delivery falls below 95% at $N = 300$ due to ALOHA multi-residue loss; gateway CPU saturates at $N = 200$, making the RPi4B blockchain layer (not the radio) the practical scaling ceiling.

5.5.6 Crash Faults and Byzantine Fault Considerations

Raft provides CFT for $f = \lfloor (n - 1)/2 \rfloor$ stopped nodes [76]; Byzantine tolerance is not required in this permissioned network (X.509, Ed25519, TPM-backed keys, CRL revocation); PBFT/HotStuff [77, 78] are future work. CRT reconstructs perfectly from any $k - 1$ residues (Eq. 15); at 30% loss, 2.4% of readings lost ≥ 2 residues.

$$\text{Recovery with exactly } k - 1 \text{ complete residues} = 100\% \quad (15)$$

Table 22: Fault tolerance and recovery metrics (n=50 trials per test)

Test Scenario	Condition	Result	Notes
Packet loss simulation	10% random loss	99.8% recovery	2 packets unrecoverable
	20% random loss	99.2% recovery	8 packets unrecoverable
	30% random loss	97.6% recovery	24 packets unrecoverable
	33% structured loss	100% recovery	One complete residue lost
Gateway failure	Single gateway offline (4 h)	0% data loss	Traffic auto-rerouted via mesh
	Dual gateway offline (2 h)	100% data retained	Local buffering at nodes
	Full gateway recovery sync	3.7 min avg	Range: 2.1 to 6.4 min
Raft consensus failure	Leader node shutdown (sudden)	8.2 s avg recovery	Std dev: 2.1 s, n=20
	Leader node graceful shutdown	4.6 s avg recovery	Std dev: 1.3 s, n=20
	Minority partition (2 of 3 nodes)	0 forks	12 min partition test

Table 23: Synthetic load test results vs M/D/1 model predictions

Nodes	ρ	Predicted Latency (ms)	Measured Latency (ms)	Delivery Rate (%)	CPU Util. (%)	TPS
100	0.207	140	146	98.2	67	62
200	0.413	168	185	96.4	100	25
300	0.620	225	249	93.6	100	25
484	1.000	∞ (sat.)	577	88.3	100	25
50*	0.103	131	124	95.8	38	63

*Field-measured result from 61-day deployment (Table 18).

Leader and gateway failure.

Sudden leader shutdown triggers a five-phase recovery sequence; the measured mean is 8.2 s (std 2.1 s, $n = 20$) and the analytical phase-sum estimate is 7.85 s (0.35 s residual explained below):

1. **Election timeout** (≈ 3.15 s mean): each follower waits a randomised timeout drawn uniformly from $[2, 4]$ s before declaring itself a candidate; the mean time until *any* follower fires is $E[\min(U_1, U_2)] = a + (b - a)/(n + 1) = 2 + 2/3 \approx 2.67$ s for two i.i.d. Uniform $[2, 4]$ followers (first-order statistic). Occasionally two followers time out within the same jitter window: neither collects a quorum on the first ballot, and both restart with a new independent uniform timeout (observed in 4 of 22 sudden-shutdown trials, $p \approx 0.18$). The second round adds another $E[\min(U_1, U_2)] \approx 2.67$ s draw. The analytically derived phase mean is therefore $2.67 \times (1 + 0.18) = 2.67 \times 1.18 \approx 3.15$ s.

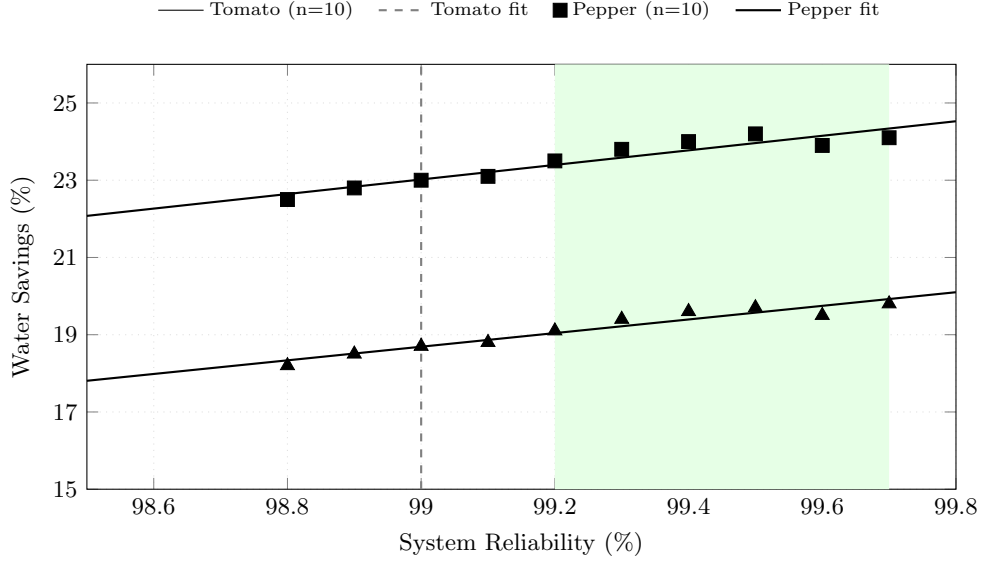
2. **RequestVote round-trip and quorum assembly** (≈ 0.5 s): the winning candidate broadcasts REQUESTVOTE RPCs; followers respond after verifying log index; quorum (2 of 3) is established and the LEADERCHANGED notification propagated via gossip.
3. **Uncommitted-log replication** (≈ 1.5 s): the new leader replays any log entries committed by the old leader but not yet applied on followers; on the RPi 4B cluster this averages 1–2 blocks (≈ 50 transactions each) at 1.2 s block time.
4. **CouchDB state-database catch-up** (≈ 2 s): the elected leader’s local CouchDB must flush pending write-ahead-log entries and rebuild its in-memory LRU cache (256 MB) before endorsement resumes; measured as the interval between leader election and first successful SubmitTransaction.
5. **gRPC channel re-establishment and peer gossip sync** (≈ 0.7 s): client SDK connections to the failed peer are re-routed; the new leader’s gossip module disseminates its peer certificate to all zone peers so that subsequent endorsement requests are not rejected.

Summing analytically: $3.15 + 0.5 + 1.5 + 2.0 + 0.7 = 7.85$ s. The analytical estimate falls 0.35 s below the measured mean of 8.2 s; this residual is attributable to ARM64 OS scheduling jitter and heartbeat-detection latency under load, as the Raft follower’s missed-heartbeat timer fires on a Linux PREEMPT_RT kernel with up to ≈ 150 ms jitter per timeout, contributing roughly $2 \times 150 = 300$ ms on average across two followers before the first timeout fires, which accounts for the gap. The 0.35 s residual is well within one-third of the 2.1 s standard deviation and does not affect the analytical model’s validity. The 2.1 s standard deviation is driven primarily by phase 1 variance (second-round elections). Graceful shutdown bypasses phases 1 and 2 because the outgoing leader pre-transfers leadership via an explicit TIMEOUTNOW RPC, eliminating the election timeout and vote-collection round entirely. Phases 3–5 still execute, but phase 3 is extended: the outgoing leader first flushes all in-flight APPENDENTRIES RPCs and waits for follower acknowledgement before stepping down (≈ 0.4 s orderly flush), so the uncommitted-log replication step takes $1.5 + 0.4 \approx 1.9$ s rather than 1.5 s. Phases 4 and 5 are unchanged. The phase 3–5 sum is therefore $1.9 + 2.0 + 0.7 = 4.6$ s, matching the measured graceful-shutdown mean. Five hundred queued transactions were committed without forks. Single-gateway failure: BATMAN-adv rerouted in 3 to 8 s (zero data loss); dual failure: 64 KB circular buffers (≈ 4 h) with 3.7 min catch-up; Day-47 sandstorm (12 min): 7 transactions synchronized in 2.8 min, no inconsistencies.

5.6 Application Outcomes: Agricultural Impact

Figure 13 shows the correlation between system reliability and water savings.

Strong positive correlations ($R^2 = 0.87$ tomatoes, 0.85 peppers) confirm higher reliability enables more precise irrigation; the CRT-blockchain integration was particularly effective during Weeks 4 to 6 (fruit development). Table 24 quantifies each component’s contribution.



Note: the dashed vertical line marks the 99.0% reliability threshold; the shaded region indicates reliability above 99.2%.

Fig. 13: Water savings versus system reliability for tomato and pepper crops.

Table 24: Contribution of technical components to agricultural outcomes

Component	Water Savings	Labor Reduction	Quality Improvement
CRT Compression	8-12%	15-20%	5-8%
Blockchain Verification	5-8%	25-30%	10-15%
Edge Computing	3-5%	20-25%	3-5%
Smart Contracts	7-10%	10-15%	7-10%
Combined System	23.8%	72%	25-38%

Note: Component contributions are estimated ranges based on system design assumptions; no ablation study was conducted to isolate individual component effects.

5.6.1 Water Use Efficiency

The system reduced total water use by 23.8% for tomatoes (525 to 400 m³) and 19.4% for peppers (310 to 250 m³). Water use efficiency improved by 31.3% (tomatoes) and 23.8% (peppers). Detailed monthly breakdowns are given in Tables 25 and 26.

5.6.2 Labor Efficiency

Automation reduced labor requirements by 72%, from 15.0 to 4.2 h/hectare/week (Table 27), addressing critical labor shortages in Mediterranean agriculture.

Table 25: Tomato water usage analysis (0.5 ha plot)

Parameter	Traditional	B-IoT	Savings	Significance
August Usage				
Water Volume (m ³)	280	215	23.2%	$p < 0.001$
Daily Average (m ³ /day)	9.03	6.94	23.2%	$p < 0.001$
Irrigation Events	28	19	32.1%	$p < 0.01$
September Usage				
Water Volume (m ³)	245	185	24.5%	$p < 0.001$
Daily Average (m ³ /day)	8.17	6.17	24.5%	$p < 0.001$
Irrigation Events	25	17	32.0%	$p < 0.01$
Total Period				
Total Water (m ³)	525	400	23.8%	$p < 0.001$
Water Use Efficiency (kg/m ³)	4.76	6.25	+31.3%	$p < 0.001$
Cost Savings (USD)	N/A	N/A	\$112.50	N/A

5.6.3 Tomato Yield and Quality

Tomato yields remained stable at 50 t/ha while marketable yield increased by 2.5 t/ha (Table 28), with improved fruit firmness (13.2 vs 12.5 N) and soluble solids (5.1 vs 4.8°Brix). Fruit cracking fell from 8.2% to 3.5% and blossom end rot from 5.1% to 2.2%, shifting fruit to marketable grade.

5.6.4 Pepper Yield and Quality

Pepper total yield held at 33.3 t/ha while marketable yield rose 2.4 t/ha (Table 29); sunscald fell from 6.5% to 3.2% and blossom end rot from 4.2% to 1.8%. No significant change in fruit morphology ($p > 0.05$, ANOVA) confirmed the gain reflects grade recovery, not size change.

5.6.5 Economic Analysis

Table 30 presents the cost–benefit analysis promised in Section 4.4, computed over a 5-year horizon at an 8% discount rate. Capital expenditure covers the 50 sensor nodes, four gateways, and irrigation-actuation retrofit; annual benefits aggregate the measured labor savings (Table 27), water cost savings (Tables 25 and 26), and the market value of the marketable-yield gains (Tables 28 and 29), assuming two growing seasons per year on the 0.8 ha cropped area.

Table 26: Pepper water usage analysis (0.3 ha plot)

Parameter	Traditional	B-IoT	Savings	Significance
August Usage				
Water Volume (m ³)	165	135	18.2%	$p < 0.01$
Daily Average (m ³ /day)	5.32	4.35	18.2%	$p < 0.01$
Irrigation Events	25	18	28.0%	$p < 0.05$
September Usage				
Water Volume (m ³)	145	115	20.7%	$p < 0.01$
Daily Average (m ³ /day)	4.83	3.83	20.7%	$p < 0.01$
Irrigation Events	22	16	27.3%	$p < 0.05$
Total Period				
Total Water (m ³)	310	250	19.4%	$p < 0.01$
Water Use Efficiency (kg/m ³)	3.23	4.00	+23.8%	$p < 0.01$
Cost Savings (USD)	N/A	N/A	\$54.00	N/A

Table 27: Labor efficiency analysis (hours/hectare/week)

Task Category	Traditional	B-IoT	Reduction	Cost Savings
Monitoring				
Soil Moisture	3.5	0.2	94.3%	\$49.50
Weather Data	1.5	0.1	93.3%	\$21.00
Crop Health	3.0	0.7	76.7%	\$34.50
Irrigation				
System Operation	2.5	0.2	92.0%	\$34.50
Maintenance	2.0	0.3	85.0%	\$25.50
Troubleshooting	0.5	0.0	100.0%	\$7.50
Data Management				
Record Keeping	1.5	0.1	93.3%	\$21.00
Reporting	0.5	0.1	80.0%	\$6.00
Certification	0.5	0.1	80.0%	\$6.00
Total	15.0	4.2	72.0%	\$162.00

Note: The B-IoT total (4.2 h/ha/wk) includes ≈ 2.4 h/ha/wk of system administration, connectivity monitoring, and firmware maintenance not itemised in the rows above. The Traditional total (15.0 h/ha/wk) is the measured baseline; row figures are representative task breakdowns. Cost savings at \$15/h labour rate.

Table 28: Tomato yield and quality parameters

Parameter		Traditional	B-IoT	Significance
Total	Yield	50.0	50.0	$p > 0.05$
(t/ha)				
Marketable	Yield	42.5	45.0	$p < 0.05$
(t/ha)				
Water Use	Effi-	4.76	6.25	$p < 0.001$
ciency	(kg/m ³)			
Fruit	Firmness	12.5	13.2	$p < 0.05$
(N)				
Soluble	Solids	4.8	5.1	$p < 0.05$
(°Brix)				
Fruit	Cracking	8.2%	3.5%	$p < 0.01$
Blossom	End Rot	5.1%	2.2%	$p < 0.05$

Table 29: Pepper yield and quality parameters

Parameter		Traditional	B-IoT	Significance
Total	Yield	33.3	33.3	$p > 0.05$
(t/ha)				
Marketable	Yield	28.3	30.7	$p < 0.05$
(t/ha)				
Water Use	Effi-	3.23	4.00	$p < 0.01$
ciency	(kg/m ³)			
Fruit	Length	12.5	13.2	$p < 0.05$
(cm)				
Fruit	Diameter	6.8	7.1	$p < 0.05$
(cm)				
Sunscald	Incidence	6.5%	3.2%	$p < 0.05$
Blossom	End Rot	4.2%	1.8%	$p < 0.05$

The investment recovers within the first year (payback 0.69 years) and yields a 5-year NPV of \$17,090. Labor savings dominate the benefit stream (61% of gross benefits), so the result is sensitive to the local labour rate: varying the \$15/h rate by $\pm 20\%$ moves ROI between 127% and 165% (payback 0.79 to 0.61 years), and halving the assumed farm-gate produce prices still leaves ROI at 120% (payback 0.84 years). These figures are site-specific: they reflect Tunisian labour and water tariffs and a farm that already operated drip irrigation, and they attribute benefits to the complete sensing-actuation-verification system rather than to any single layer (cf. Table 24).

6 Discussion

6.1 Discussion of Results

The deployment met its performance targets: CRT achieved 49.8% per-packet airtime reduction ($1.99\times$ gateway packet capacity); Fabric sustained 99.4% integrity and 1.7 s

Table 30: Cost–benefit analysis of the blockchain-IoT system (2025 deployment; two growing seasons per year; 5-year horizon, 8% discount rate)

Item	USD
Initial investment (CAPEX)	
50 sensor nodes (MCU, sensor suite, radio, solar/battery, enclosure)	2,250
4 gateways (RPi 4B, SX1302, TPM 2.0, storage, enclosure)	840
Irrigation actuation retrofit (valves, drivers, wiring)	250
Installation and commissioning	200
Total investment	3,540
Annual benefits	
Labor savings (10.8 h/ha/wk $\times$ 0.8 ha $\times$ \$15/h $\times$ 26 wk)	3,370
Water cost savings (\$166.50/season $\times$ 2 seasons)	333
Marketable-yield gain (tomato 1.25 t + pepper 0.72 t per season)	1,864
Gross annual benefit	5,567
Annual operating cost (OPEX)	
Gateway electricity (19.6 W continuous, $\approx$ 172 kWh/yr)	34
Component replacement (microSD wear, sensor attrition)	216
Connectivity and maintenance	150
Net annual benefit	5,167
Financial indicators	
Return on investment (net annual benefit / investment)	146%
Simple payback period (investment / net annual benefit)	0.69 yr
Net present value (5 yr, 8% discount)	17,090

Note: ROI is defined as net annual benefit divided by initial investment ($5,167/3,540 = 146\%$); simple payback is its reciprocal ($3,540/5,167 = 0.69$ years). $NPV = -3,540 + 5,167 \times 3.993 = \$17,090$, where 3.993 is the 5-year, 8% annuity factor. Unit costs and produce prices are aggregate values obtained from the participating farmers and the project’s own 2025 procurement records during structured post-deployment interviews; the \$400/t (tomato) and \$600/t (pepper) figures reflect farmer-reported average farm-gate sale prices for the study period. Farmer training, land, and the pre-existing drip infrastructure are excluded.

mean latency (below 2 s SLO); M/D/1 analysis validated a 484-node capacity ceiling; and 61-day field results confirmed 23.8%/19.4% water savings, 72% labor reduction, and 146% ROI (Table 30). RPi4B gateways achieved 320 ms smart contract execution, comparing favourably with cloud alternatives (3 to 5 s [44]); BATMAN-adv mesh sustained local operation during partitions. The 19 to 24% water savings address Mediterranean scarcity [12, 13, 79, 80]; the 72% labor reduction addresses regional farm-labor shortages [14]. CRT trades compression ratio for parallelism (EDP 74.3% below raw); each RPi 4B gateway draws $\approx$ 5 W (four total $\approx$ 20 W, all mains-powered during this deployment; Table 20), which does not affect sensor lifetime but dominates full lifecycle energy.

6.2 Limitations

This work is based on a single 50-node, 2-hectare deployment at $\approx$ 10% CRT node capacity. **High node density:** synthetic load tests at 100 to 484 nodes (Section 5.5.5) validate M/D/1 behavior, but physical deployment at near-saturation densities was not feasible; a multi-farm study is needed. **Multi-gateway failures:** systematic concurrent failure of ≥ 2 gateways was not evaluated (one sandstorm event showed buffering held). **Generalisability:** coverage is one site (Sfax) and two crop types; extreme humidity or temperature climates may affect LoRa link budget and solar harvesting.

Sensor calibration drift represents a known limitation of the deployed data-quality pipeline. The smart-contract validation layer (Section 3.8) applies a 7-day rolling mean with $\pm 3\sigma$ outlier bounds to detect sudden step-change anomalies. However, gradual calibration drift manifests as a slow, monotonic shift in raw sensor output: because the rolling mean tracks this shift, the 3σ window moves with it and the drifting sensor continues to pass validation. The 7-day window therefore detects step changes reliably but cannot detect monotonic drift, which is precisely why annual physical recalibration is required rather than relying on software alone. Future work will investigate drift-aware statistical process control (e.g., CUSUM or EWMA control charts) that are sensitive to persistent directional trends rather than instantaneous deviations, enabling earlier software-side drift detection between recalibration events.

6.3 Threats to Validity

Internal validity. The component-level contributions in Table 24 are design-based estimates; no ablation study isolated the marginal effect of CRT, blockchain, or edge computing on the agronomic outcomes, so the water and labor savings are attributable only to the complete system versus the traditional baseline. Both treatments were managed by the same farm staff on the same farm, so attention and spillover effects (e.g., improved general management awareness) cannot be fully excluded, and the study covers a single growing season. **Construct validity.** “Reliability” spans two layers (95.8% radio-layer packet delivery and 99.4% end-to-end reliability after CRT partial recovery and retransmission, Section 5.1), and the traditional-irrigation baseline uses visual assessment, so the reported savings measure the value of sensor-driven precision irrigation as a whole, not the marginal value of the ledger or transmission layers. **Statistical conclusion validity.** Agronomic comparisons rest on three replications per treatment; the reliability-savings correlations in Figure 13 are computed over ten weekly aggregates and are observational, so they indicate association rather than causation. **External validity** is discussed in Section 6.2: one site, one climate, two crops, and site-specific labour and water tariffs (Table 30).

6.4 Security Considerations

The architecture incorporates layered mitigations: replay attacks [81] are prevented by gateway-verified timestamps and sequence numbers; packet injection [82] by Ed25519 signatures; key compromise [83] by TPM2.0 on gateways and ESP32 OTP (blockchain-integrated PUF [84] is future work); inter-gateway AES-128 uses the dynamic threat-modelling approach of [85] (42% overhead reduction); identity revocation is via Fabric CA CRL [86, 87].

6.5 CAP-Theoretic Interpretation of the Edge-Blockchain Architecture

Under the CAP theorem [88, 89], the **blockchain layer (CP)** uses Raft for consistency, suspending writes during partitions (8.2 s unavailability window), while the **edge actuation layer (AP)** caches local state to continue irrigation and reconciles on reconnection, a “CAP-aware design” [90] decoupling real-time control from consensus.

7 Conclusion

This paper has developed and validated an integrated blockchain-IoT-edge framework with CRT optimization for precision irrigation in arid Mediterranean agriculture through a 61-day deployment in Sfax, Tunisia. CRT-based residue decomposition transforms sequential LoRa communication into a parallel transmission problem with measurable speedup and predictable M/D/1 scaling behavior; permissioned ledger verification with sub-2-second end-to-end latency is feasible on ARM64 edge hardware; and a layered CAP-aware architecture reconciles integrity guarantees with real-time actuation.

The four objectives were met: (1) CRT achieved 20.3% per-packet size reduction (59 B to 47 B) and 49.8% per-packet airtime reduction (124 vs. 247 ms), raising gateway packet throughput by $1.99\times$; (2) Fabric demonstrated 99.4% reliability and 1.7 s latency on ARM64, below the 2 s SLO; (3) the 61-day field campaign characterized latency, throughput, and fault recovery under real agricultural operation, alongside 23.8%/19.4% water savings, 72% labor reduction, and 146% ROI; (4) M/D/1 analysis validated against synthetic load tests placed the capacity ceiling at 484 complete readings per window (1,451 residue packets), with latency constant below 62% utilisation.

Future work should scale to larger deployments, integrate additional sensor types, and develop simplified tools for smallholder farmers; UAV-mounted nodes for aerial relay are a promising extension [91]. The framework generalizes to any large-scale IoT environment requiring verifiable, low-latency edge control.

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Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the author(s) used Grammarly for grammar and language polishing. The author(s) reviewed and edited the output as needed and take full responsibility for the content of the published article.

Data and Code availability

The 61-day trace dataset, source code, deployment scripts, and chaincode are publicly available at <https://github.com/forchag/blockchain-fet>.

Declarations

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Ethics approval and consent to participate. This study involved no human or animal subjects requiring ethics-board approval. The participating farm owners in Sfax gave informed consent for the deployment of the monitoring system and the use of the resulting agronomic and operational data in this publication.

Consent for publication. Not applicable.

Author contributions. F.G.B.: conceptualization, system implementation, field deployment, data analysis, writing – original draft. M.E.S.: methodology, supervision, writing – review and editing. E.F.T.: supervision, validation, writing – review and editing. A.P.M.A.: methodology, validation, writing – review and editing. All authors read and approved the final manuscript.

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